Isifundo sezibalo sesithwalise abafundi base Ningizimu Afrika ubunzima iskhathi eside , Iningi labafundi laze laphonsa ithawula ngemfundo ngenxa yokuhluleka ukuqonda lesi sifundo .

Sibe sesithatha ithuba lokuba sihlolisise loludaba sase siqonda ukuba ayikho into esemqoka kumfundi onsundu njengo limi lwasekhaya okuyinto eyigugu kakhulu kubona . Izincwadi eziningi zezibalo zibhalwe ngolimi okungasilo olwase Afrika kanti futhi zibhalwe ngendlela yokuba azimniki umfundi onsundu umsuka wesifundo , Zifuna okuba umfundi asebenzise lezi zincwadi uma esenolwazi oluphelele ngesifundo esedinga okuba aziqeqeshe ukuze akwazi ukubhala kahle izivivinyo zakhe .

Abafundi abaningi banenkinga yokuba umyalezo osuke ushiwo u Thisha e kilasini bayawuzwa kodwa uma sebefika emakhaya sebewukhohliwe , Uma sebedinga ukuligubha lolulwazi besebenzisa izincwadi bathole ukuba lolulwazi lubhalwe ngolimu okungasilo olwabo futhi abaluqondi kahle .

Ulimi kanye nomsuka wesifundo izinto ezisemqoka ezenza kubelukhuni kubafundi ukuqonda kahle izibalo . Sibe sesihlanganisa ulwazi esinalo ukwenza incwadi yezibalo yesi Zulu ezosiza abafundi abansundu ukuba baqonde izibalo ngolimi lwabo .

Esinethemba lokuba izomsiza kakhulu umfundi uma esefika khaya esedinga ukubheka ulwazi aselukhohliwe obelushiwo u thisha e kilasini.

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Chapter 1

Sequences And Series

Arithmetic Sequence

Emabangeni adlule isifundo samaphethini bekuyisifundo ebesifundwa ngokuphindelela , Kepha kulelibanga likamatikuletsheni sizofunda nge phethini ebizwa nge Arithmetic Sequence.

I-Arithmetic sequence uhlobo lwephethini olusebenzisa izimpawu ezimbili , okuwuphawu lokususa(-) kanye nophawu lokuhlanganisa(+) .Sisebenzisa uphawu lokususa ukuthola ukuba izinombolo zethu zishiyana ngebanga elingakanani siphinde sisebenzise uphawu lokuhlanganisa ukuthola inombolo elandelayo kwi phethini uma sesitholile ukuba zishiyana ngaliphi ibanga. Uhlobo lwefomula esijwayele ukulisebenzisa yilolu :

$$*T_n = a + d(n - 1)$$

* T_n(usitshela ukuba inombolo/ i-themu yesingaki)i.e T_1 = inombolo/i-themu yokuqala. *-Tn uhlezi unenani (value) .i.e $T_1 = 8$*

* a(inombolo yokuqala kwiphethini yethu).

* d(kusitshela ukuba izinombolo zethu zishiyana ngebanga elingakanani okujwayeleke ukuba kubizwe nge *COMMON DIFFERENCE*).Ukuthola ukuba izinombolo zishiyana ngebanga elingakanani sisebenzise le formula :

$$\mathbf{d = T_n - T_{n-1} \text{ noma } T_{n_1} - T_{n_1 - 1} = T_{n_2} - T_{n_2 - 1} = d}$$

Lapho sithatha I nombolo engaphambili kwi phethini sisusa I nombolo engemumva ukuthola ukuba zishiyana ngaliph ibanga.

*n(kusitshela ukuba inombolo yethu ikwisikhundla/(position) yesingaki).

IZIBONELO :

*Consider the linear pattern : 5 ; 9 ; 13, …

Determine the Common Difference

$d = T_n - T_{n-1}$

$d = T_3 - T_{3-1}$ (T_3 umele inombolo yesithathu kwi phethini)

$d = 13 - 9 = 4$

Noma

*$T_n - T_{n-1} = T_n - T_{n-1} = d$

$T_3 - T_{3-1} = T_2 - T_{2-1} = d$

$T_3 - T_2 = T_2 - T_1 = d$

$13 - 9 = 9 - 5 = d$

$4 = 4 = d$ (kwi Arithmetic sequence I banga izinombolo ezishiyana ngalo lihlezi lilingana)

Determine the 4th Term

$T_n = a + d(n-1)$

$T_4 = 5 + 4(4-1)$ …$n = 4$(I nombolo/i-themu yesine ikwisikhundla/position sesine)

$T_4 = 17$

$T_5 = 17 + 4 = 21$

Determine the General Term of the sequence

$T_n = 5 + 4(n-1)$ … I- Fomula ye phethini yethu

$3x + 1$; $2x$; $3x - 7$ are the first three terms of an arithmetic sequence.

Calculate the value of x.

$T_n - T_{n-1} = T_n - T_{n-1} = d$

$T_3 - T_2 = T_2 - T_1 = d$

$3x - 7 - (2x) = 2x - (3x + 1)$

$3x - 7 - 2x = 2x - 3x - 1$

$x - 7 = -x - 1$

$2x = 6$

$x = 3$

Amanothi okumele ahlolisiswe

Uma izinombolo zethu kwiphethini zingakabi inombolo eliqiniso /ejwayelekile(izinombolo zisaxubene nama alfabhethi). Kubalulekile ukusebenzisa le formula :

$T_n - T_{n-1} = T_n - T_{n-1} = d$

ukuthola ukuba leyo alfabhethi iyiyiphi I nombolo mase uthola izinombolo zakho zeqiniso.

Determine the Common Difference

$3(3) + 1$; $2(3)$; $3(3)-7$; …

10 ; 6 ; 2

$d = T_n - T_{n-1}$

$d = T_3 - T_2$ noma $d = T_2 - T_1$

$d = 2 - 6 = -4$ $d = 6 - 10 = -4$

Determine the General Term of the Sequence

$T_n = 10 - 4(n - 1)$

Given T_4 and T_{14} of an Arithmetic Sequence whereby $T_4 = 16$ and $T_{14} = 86$. Determine the 1_{st} term and the common difference.

$T_4 = 16$	$T_{14} = 86$
$a + d(n - 1) = 16$	$a + d(n - 1) = 86$
$a + d(4 - 1) = 16$	$a + d(14 - 1) = 86$
$a + 3d = 16$	$a + 13d = 86$
$d = \frac{16 - a}{3}$ …(1)	$d = \frac{86 - a}{13}$ …(2)

$$(1) \ldots (2)$$

$$\frac{16 - a}{3} = \frac{86 - a}{13}$$

$$13(16 - a) = 3(86 - a)$$

$$208 - 13a = 258 - 3a$$

$$208 - 258 = 13a - 3a$$

$$-50 = 10a$$

$$-5 = a$$

$$d = \frac{16 - (-5)}{3} = \frac{21}{3} = 7$$

$-5 ; 2 ; 9 ; 16 ; 23 ; 30 , 37 ; \ldots$

$T_n = -5 + 7(n - 1)$

Uma unikezwe ukuba leyonombolo/leyo-themu (T_n) eyesingaki waphinde wanikezwa nenani(value) yayo I nombolo. i.e Tn = 9

***qala u bale u Tn ukuba uwubani (Tn wethu uyalingana nenombolo abasinikeze yona)**

***bese wenza I alfabhethi eyodwa kwi formula yethu(a or d) ilingane ne nani /wenze alfabhethi eyodwa ibe I <u>subject of the formula.</u>**

***Enza womabili ama themu(Tn) bese uwabala/solve ngokulingana. (<u>Solve Simultaneously</u>).**

UMSEBENZI WASEKHAYA

1) Determine the common difference and the general term of the following arithmetic sequences.

a) -8 ; -2 ; 4 ; 10 …

b) y ; 17 ; 26 ;35 ; …

c) $2p + 3$; $p + 6$; $p - 2$; …

d) 7 ; x ; y ; -11 ; …

2) Given the Arithmetic Series : 18 + 24 + 30 + … +300

a) Determine the number of terms in this series.

3) Given the arithmetic sequence : 35 , 28 ; 21 ; …

a)Calculate which term of the sequence will have a value of -140.

4) Given the following arithmetic term values , Calculate the first term and the common difference

a) $T_5 = 27$ and $T_{14} = 108$

b) $T_2 = 19$ and $T_6 = 87$

c) $T_3 = 43$ and $T_{10} = 99$

d) $T_5 = 78$ and $T_{12} = 113$

5) The first three terms of an arithmetic sequence are :

$2z – 1$;$4z – 1$; $6z – 1$.

a)Determine T_{30} in terms of z .

Sum Of An Arithmetic Sequence

Kulesi sihloko yilapho sisuke sesigxile ekuhlanganiseni zonke izinombolo ezikwi phethini yethu , Kepha ukuhlanganisa izinombolo nga yinye / ngayodwana kwenza umsebenzi ube nzima . Yingakho kwafikwa ne fomula yokuhlanganisa zonke izinombolo ngesikhathi esisodwa ukwenza umsebenzi ube lula . I fomula yokuhlanganisa izinombolo ezikwi phethini I Arithmetic Sequence :

$$S_n = \frac{n}{2}(a + L) \quad \text{noma} \quad S_n = \frac{n}{2}[2a + d(n - 1)]$$

* S_n (umele isamba sazo zonke izinombolo uma sezihlanganisiwe).

* d(ibanga izinombolo zethu ezihlukaniseke ngalo kwiphethini).

* n (isamba sezikhundla/ position ezikhona kwi phethini yethu. Kusitshela okuba zingaki izinombolo ezikhona kwi phethini yethu).

* a (inombolo yokuqala kwiphethini yethu).

* L(inombolo yokugcina kwi phethini yethu).

Okubalulekile

I formula engasesandleni sokunxele siyisebenzisa uma siyazi I nombolo yokugcina kwi phethini yethu kanti engakwisandla sokudla siyisebenzisa uma singenayo inombolo yokugcina kepha siyayazi I Common difference (d).

IZIBONELO :

Given the arithmetic series : 18 + 24 + 30 + … + 300

a)Determine the number of terms of this series.

$T_n = a + d(n - 1)$

$300 = 18 + 6(n - 1)$

$300 = 18 + 6n - 6$

$6n = 288$

$n = 48$

b) Calculate the sum of this series.

$S_n = \frac{n}{2}[2a + d(n - 1)]$

$= \frac{48}{2}[2(18) + 6(47)]$

$= 7632$

noma

$S_n = \frac{n}{2}(a + L)$

$= \frac{48}{2}(18 + 300)$

$= 7632$

c) Calculate the sum of all the whole numbers up to and including 300 that are not divisible by 6.

$S_{1to300} = \frac{300}{2}[2(1) + 1(299)]$

$= \frac{300(301)}{2}$

$= 45150$

sum not divisible by 6 is:

$= 45150 - (7632 + 6 + 12)$

$= 37500$

Given the following arithmetic sequence : 13 ; 8 ; 3 ; …

a)Determine the value of the 50^{th} term.

$a = 13$ and $d = -5$

$T_n = a + d(n - 1)$

$T_{50} = 13 + (50 - 1)(-5)$

$T_{50} = -232$

b)Calculate the sum of the first fifty terms.

$S_n = \frac{n}{2}[2a + d(n - 1)]$ noma $S_n = \frac{n}{2}(a + L)$

$S_{50} = \frac{50}{2}[2(13) + (50-1)(-5)]$ $S_{50} = \frac{50}{2}(13 + -232)$

$S_{50} = -5475$ $S_{50} = 25(-219) = -5475$

Okubalulekile :

Ukuze ukwazi ukubala isamba sezinombolo zakho ezikwiphethini udinga ukuhlolisisa ulwazi onikezwe lona , uma unikeziwe I nombolo yokugcina kwi phethini sebenzisa le formula:

$$S_n = \frac{n}{2}(a + L)$$

Uma unganikezwanga sebenzisa le formula :

$$S_n = \frac{n}{2}[2a + d(n - 1)]$$

UMSEBENZI WASEKHAYA

1) Consider the linear pattern : 5 ; 7 ; 9; …

a)Determine the T_{51}

b) Calculate the sum of the 51 terms

2) The following is an arithmetic sequence : $2x - 4$; $5x$; $7x - 4$

a) Determine the value of x

b) Determine the first 3 terms

c) Determine the sum of the first 60 tems

3) How many terms are there in the following sequence?

17 ; 14 ; 11 ;8 ; … ; - 2785

a) Calculate the sum of the first 935 terms

4) Consider the arithmetic sequence :

-8 ; -2 ; 4 ; 10 ; ….

a)Write down the next two terms of the sequence.

b) if the n^{th} term of sequence is 148, determine the value of n.

c)Calculate the smallest value of n for which the sum of the first n terms of the sequences will be greater than 10 140.

Sigma Notation

I Sigma Notation isihloko esibheke ekuthatheni iphethini esibhaliwe isibhale ngendlela e lula, okwenza kubelula ukuthola isamba sazozonke izinombolo kwi phethini(S_n) kanti iphinde inciphise ukubhaleka kwezinombolo ze phethini ngobuningi . Kwi sigma notation sike sisebenzise lolu hlobo lwe fomula :

Inani Iesikhundla sokugcina kwiphethini yethu

$$\sum_{k=1}^{n} Tn$$

I Fomula ye phethini yethu

Kumele isamba sezinombolo ze phethini (Sum)

Inani lesikhundla sokuqala kwi phethini yethu(1st position)

*Inani = value

*Isikhundla = position

*Ukuthola ukuba iphethini yethu inezinombolo ezingaki / ama-terms amangaki /izikhundla ezingaki uma sibuka i sigma notation yethu sithatha **inani lesikhundla sokugcina** sisuse**(-)** kulo **inani lesikhundla sokuqala** bese inombolo esiyitholayo siyihlanganisa no 1.   Isibonelo : $\sum_{k=4}^{10}(2k+4)$

k = 4 (inani lesikhundla sokuqala kwiphethini yethu)

10 (inani lesikhundla sokugcina kwiphethini yethu)

n(ukuthola isamba sezikhundla zethu) = (10 – 4) + 1 = 7

*Uma sesiqede ukubalela inani elikwisikhundla sokuqala siyaqhubeka sibalele inani elilandelayo(k odlule +1) size sigcine ukubala uma sesibalele inani lesikhundla sokugcina. i.e [2(4)+4] +[2(5)+4] + [2(6)+4] +[2(7)+4] +[2(8)+4]+[2(9)+4]+[2(10)+4]

inani lokugcina **inani lokuqala**

IZIBONELO(Arithmetic Sequence Sigma):

Given the following Sigma Notation : $\sum_{n=4}^{70}(2n-4)$

$2(4) - 4 + 2(5) - 4 + 2(6) - 4 + \ldots$

$4 + 6 + 8$

The Sequence is 4 ; 6 ; 8 … This is an arithmetic series

$d = T_2 - T_1 = 2$

$n = (70 - 4) + 1 = 67$

$S_n = \frac{n}{2}[2a + d(n-1)]$

$S_{67} = \frac{67}{2}[2(4) + 2(67 - 1)]$

$S_{67} = 33.5\,[8 + 132] = 4690$

*So the sum of the following Sigma Notation is <u>4690</u>.

Calculate : $\sum_{k=1}^{30}(3k+5)$

$[3(1) + 5] + [3(2) + 5] + [3(3) + 5] + \ldots$

$8 + 11 + 14 + \ldots$

$d = T_2 - T_1 = 3$

$n = (30 - 1) + 1 = 30$

$S_{30} = \frac{30}{2}[2(8) + 3(30 - 1)] = 15[16 + 87]$

$= 1545$

<u>Okubalulekile</u>

***uma unikezwe I Sigma Notation kubalulekile ukuqala uyehlukanise ukuze uthole amanani ephethini yakho bese uyabheka ukuba i phethini yakho iyuhloboluni, bese usebenzisa I formula yalolo hlobo yokubala isamba (I phethini ingaba Arithmetic or Geometric)**

Geometric izofundwa kwisifundo esilandelayo

Geometric Series

I- Geometric Series uhlobo lwephethini olusebenzisa izimpawu ezimbili, okuwuphawu loku *multiplier* (×) kanye nophawu loku *divide* (÷) .Sisebenzisa uphawu loku *divide* ukuthola ukuba izinombolo zethu zishiyana ngebanga elingakanani siphinde sisebenzise uphawu loku *multiplier* ukuthola inombolo elandelayo uma sesitholile ukuba zishiyana ngaliphi ibanga. Uhlobo lwefomula esijwayele ukulisebenzisa yilolu :

$$T_n = a.r^{n-1}$$

Lapho :

T_n (Isitshela ukuba inombolo/ i-themu yesingaki). i.e T_3=inombolo/i-themu yesithathu.

a (Inombolo yokuqala kwiphethini yethu).

r (Kusitshela ukuba izinombolo zethu zishiyana ngebanga lesingaki okujwayeleke ukuba kubizwe nge **COMMON RATIO(r)).**Ukuthola ukuba izinombolo zishiyana ngebanga elingakanani sisebenzise le formula :

$$r = \frac{Tn}{Tn-1} \quad \text{noma} \quad \frac{Tn_1}{Tn_1-1} = \frac{Tn_2}{Tn_2-1} = r$$

n (Kusitshela ukuba inombolo yethu ikwisikhundla/*(position)* yesingaki).

*Umehluko Phakathi kwe Geometric ne Arithmetic series ukuba I Geometric isebenzisa uphawu loku *multiplier(x)* neloku *divide(/)* kanti i-Arithmetic isebenzisa uphawu lokuhlanganisa(+) nelokususa (-) kwamaphethini .

IZIBONELO :

1) Given the following pattern : 3 ; 6 ; 12 ; …

a)Determine the Common Ratio

$r = \frac{Tn}{Tn-1}$ noma $\frac{Tn_1}{Tn_{1-1}} = \frac{Tn_2}{Tn_{2-1}} = r$

$r = \frac{T_2}{T_2-1} = \frac{T_2}{T_1}$ $\frac{T_3}{T_{3-1}} = \frac{T_2}{T_{2-1}} = r$

$r = \frac{6}{3} = 2$ $\frac{T_3}{T_2} = \frac{T_2}{T_1} = \frac{12}{6} = \frac{6}{3} = 2 = 2 = r$

b)Determine the 10th Term

$Tn = ar^{n-1}$

$T_{10} = 3(2)^{10-1} = 3(2)^9 = 3 \text{ x } 512 = 1536$

2)Determine the number of terms in the sequence : 2 ; 4 ; 8 ; … ; 1024

$r = \frac{Tn}{Tn-1}$

$r = \frac{T_2}{T_1} = \frac{4}{2} = 2$

$Tn = ar^{n-1}$

$1024 = 2(2)^{n-1}$

$512 = 2^{n-1}$

$2^9 = 2^{n-1}$

$9 = n - 1$

$10 = n$

2) If 5 ; x ; 45 are the first three terms of a geometric sequence, determine the value of x.

$$\frac{Tn_1}{Tn_{1-1}} = \frac{Tn_2}{Tn_{2-1}} = r$$

$$\frac{T_3}{T_2} = \frac{T_2}{T_1}$$

$$\frac{45}{x} = \frac{x}{5}$$

$$x = \pm\sqrt{225}$$

$$x = \pm 15 \quad \therefore x = 15$$

b) Given the geometric sequence whose 8^{th} term is 9 and whose 10^{th} term is 25, Determine the first term of the pattern.

$Tn = ar^{n-1}$ $\qquad$ $Tn = ar^{n-1}$

$T_8 = ar^{8-1}$ $\qquad$ $T_{10} = ar^{10-1}$

$ar^7 = 9 \ldots(2)$ $\qquad$ $ar^9 = 25 \ldots(1)$

$(1) \ldots (2)$

$$\frac{ar^9}{ar^7} = \frac{25}{9}$$

$$r^2 = \frac{25}{9}$$

$$r = \frac{5}{3}$$

$$a = \frac{9}{(5/3)^7} = 9 \times (3/5)^7 = \frac{1}{25}$$

UMSEBENZI WASEKHAYA

1)If $r = \frac{1}{5}$ and a = 2000 , determine :

a) Which term of the series will have a value of $\frac{6}{15\ 625}$

b) Determine the first 6 terms of the sequence

2) Given the geometric series: $z + 90 + 81 + \ldots$

a) Calculate the value of z

b) Determine the value of the next two terms of the sequence

3) Given that in a geometric sequence $T_9 = 768$ and $T_{13} = 12\ 288$. Determine the value(s) of the common ratio and the first term of the sequence

4)Given a geometric series : $256 + p + 64 - 32 + \ldots$

a)Determine the value of p.

b)Determine the values of the next 3 terms of the sequence

5)Given the following geometric series : $-2 + x + \ldots$

a)Write down the value of the third term in terms of x

6) The first two terms of a geometric sequence and an arithmetic sequence are the same. The first term is 12. The sum of the first three terms of the geometric sequence is 3 more than the sum of the first three terms of the arithmetic sequence. Determine TWO possible values for the common ratio, r, of the geometric sequence.

Sum Of A Geometric Sequence

Lesi sihloko siyafana nesihloko ebesisifunde ngaphambilini(sum of an arithmetic sequence) lapho sisuke sihlanganisa zonke izinombolo ezikwi phethini yethu ukuze sithole isamba sazo(total) , kepha okuhlukile ukuba kwi Geometric Sequence ukuze sithole isamba sayo sisebenzise I formula engafani/ehlukile ne formula yokuthola isamba se arithmetic sequence. I formula yokuthola isamba se Geometric sequence :

$$S_n = \frac{a(r^n - 1)}{r - 1} \quad \text{noma} \quad S_n = \frac{a(1 - r^n)}{1 - r}$$

Lapho :

S_n (umele isamba sazo zonke izinombolo uma sezihlanganisiwe).

a (inombolo yokuqala kwiphethini yethu).

r (ibanga izinombolo zethu ezihlukaniseke ngalo kwiphethini).

n (isamba sezikhundla/ position ezikhona kwi phethini yethu. Kusitshela okuba zingaki izinombolo ezikhona kwi phethini yethu).

*Kubalulekile ukuba uqale uhlolisise iphethini yakho ukuba iyuhlobo luni mande uzame ukwenza izibalo ezidingakalayo, ukubalela ibanga izinombolo ezishiyana ngalo waphinde wathola ukuba liyalingana ngokuqhubeka kwe phethini alishintshi, iyona yenye yezindlela yokuthola ukuba iphethini yakho iyuhloboluni.

IZIBONELO :

1)Determine 3 + 6 + 12 + 24 + … to 10 terms

$r = \frac{T_3}{T_2} = \frac{T_2}{T_1}$

$r = \frac{12}{6} = \frac{6}{3} = 2$

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$S_{10} = \frac{3(2^{10} - 1)}{2 - 1} = 3(1024 - 1)$

$S_{10} = 3069$

2) if 2 + 6 + 18 + … = 728 , determine the value of n.

$r = \frac{T_3}{T_2} = \frac{T_2}{T_1}$

$r = \frac{18}{6} = \frac{6}{2} = 3$

$S_n = \frac{a(r^n - 1)}{r - 1}$

$728 = \frac{2(3^n - 1)}{3 - 1}$

$728 = 3^n - 1$

$729 = 3^n \qquad 3^6 = 3^n \qquad n = 6$

1)Given the geometric series : $\frac{5}{8} + \frac{5}{16} + \frac{5}{32} + \ldots = F$

I)Determine the value of F if the series has 21 terms.

$$r = \frac{5}{16} \div \frac{5}{8}$$

$$r = \frac{5}{16} \times \frac{8}{5}$$

$$r = \frac{40}{80} = \frac{1}{2}$$

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$S_{21} = \left[\frac{(5/8)\,((½)^{21} - 1)}{½ - 1}\right]$$

$$S_{21} = \frac{0{,}625\,(-\,0{,}9999)}{-0{,}5}$$

$$S_{21} = 1{,}25$$

UMSEBENZI WASEKHAYA

1)Given the geometric series : $k + 90 + 81 + \ldots$

a)Calculate the value of k.

b)Show that the sum of the first n terms is $S_n = 1000(1 - (0{,}9)^n)$

2) The following geometric series is given: $y = 5 + 15 + 45 + \cdots$ to 35 terms.

a) Calculate the value of y.

3)Given the geometric sequence: 7 ; q ; 63 ; … Determine the possible value of q.

4)The first term of a geometric sequence is 15. If the second term is 10, calculate:

a) $T10$
b) $S9$

5) Given the following sequence : $16 + 8 + 4 + 2 + \cdots$

a) Determine the sum of the first forty terms of the series.

b) Write the given series in sigma notation.

6) The following sequence: 2 ; 2 ; 5 ; 4 ; 8 ; 8 ; ... is a combination of a linear and geometric sequence.

a) If the pattern continues, then write down the next TWO terms.

b) Calculate the sum of the first 40 terms of the sequence.

7)Given: $16 + 3 + 8 + 3 + 4 + 3 + 2 + \ldots$

a)Determine the sum of the first 40 terms of the series, to the nearest integer.

Sum To Infinity (Geometric Sequence)

Ngaphambilini sigxile ekufundeni ngamaphethini esisuke siwazi ukuba agcina kusiphi isikhundla(n) / kuyiphi I nombolo / kuyiphi I-term . Kepha siyakwazi ukuba ne phethini engenalo inani eligcina kulo eqhubeka njalo njalo , Ukuqhubeka kwephethini ingabi nalo inani eligcina kulo kuke kuthiwe leyo phethini I infinity okusho ukuba ayipheli. Uhlobo lwalephethini siluthola kwi Geometric sequence kanti siyakwazi ukubalela isamba sayo(sum) lapho sisebenzisa nayi I formula :

$$S\infty = \frac{a}{1 - r}$$

Lapho :

$S\infty$ (umele isamba sazo zonke izinombolo uma sezihlanganisiwe) (represents sum to infinity∞).

a (inombolo yokuqala kwiphethini yethu).

r (ibanga izinombolo zethu ezihlukaniseke ngalo kwiphethini).

Convergence of a Geometric sequence

I phethini yethu okuyi Geometric sequence I Khonveja lapho ibanga izinombolo zethu ezishiyana ngalo kwi phethini(r) lingaphezulu kuno -1 kodwa futhi lingaphansi kuno 1. Lapho sisebenzisa loluhlobo lwe fomula ukuthola ukuba I phethini yethu iya khonveja noma cha : **(-1 < r < 1).**

Ukuthola ukuthi I phethini yethu iya khonveja noma cha kulele ekutheni sithole ukuba izinombolo ezikwi phethini yethu zishiyana ngebanga elingakanani(r) . Uma lingaphezulu kuno -1 kodwa futhi lingaphansi kuno 1 lokho kusitshela okuba I phethini yethu iya khonveja. Uma inombolo yethu(r) ingaphezulu kwa 1 nom ingaphansi kwa -1 iphethini yethu ayi khonveji.

IZIBONELO :

1)Given $3 + 1 + \frac{1}{3} + \frac{1}{9} + \ldots$

a)Determine the sum to infinity.

$$\frac{T_3}{T_2} = \frac{T_2}{T_1} = 1$$

$$\frac{1}{3} \div 1 = \frac{1}{3}$$

$$\frac{1}{3} = \frac{1}{3} = r$$

$$S\infty = \frac{a}{1 - r}$$

$$S\infty = \frac{3}{1 - \frac{1}{3}}$$

$$S\infty = \frac{9}{2}$$

b)Determine if the series converges or not

Since our r is greater than -1 but less than 1 our series do converges.

2)Given the series : $3(2x-3)^2 + 3(2x-3)^3 + 3(2x-3)^4 + \ldots$ for which values of x will the series converge?

$$r = \frac{T_2}{T_1}$$

$$r = \frac{\cancel{3}(2x-3)^{\cancel{3}\,1}}{\cancel{3}(2x-3)^{\cancel{2}}}$$

$r = 2x - 3$

$-1 < 2x - 3 < 1$ (adding 3 on both sides)

$2 < 2x < 4$ (dividing by 2 on both sides)

$1 < x < 2$

Our series will converge on $1 < x < 2$

3)Given a geometric sequence : 36 ; - 18 ; 9 …

a)Calculate the sum to infinity

$$r = \frac{9}{-18} = \frac{-18}{36}$$

$$r = -\frac{1}{2}$$

$$S\infty = \frac{36}{1-(-\frac{1}{2})}$$

$S\infty = 24$

UMSEBENZI WASEKHAYA

1)Consider geometric series : $3 + d + \frac{d^2}{3} + \frac{d^3}{9} + \ldots$

a)For which value(s) of d will the series converge?

b)It is given that : $3 + d + \frac{d^2}{3} + \frac{d^3}{9} + \ldots = \frac{27}{7}$, Calculate the value of d

2) For which value(s) of k will the series:

$$\left(\frac{1-k}{5}\right) + \left(\frac{1-k}{5}\right)^2 + \left(\frac{1-k}{5}\right)^3 + \cdots \text{ converge?}$$

3) If $(a+1)+(a-1)+(2a-5)+\cdots$ are the first three terms of a convergent geometric series, calculate:

a) The value of a where $a > 0$

b)The sum to infinity of the series

4)The sum to infinity of a geometric series with positive terms is 16 and the sum of the first two terms is 12. Determine the values of a and r.

5) Given the geometric series: $9x^2 + 6x^3 + 4x^4 + \cdots$

a)For which value(s) of x will the series converge?

6) Given the geometric series: $(x-2)+(x^2-4)+(x^3+2x^2-4x-8)+\cdots$

a) Determine the value(s) of x for which the series converges.

b) If $x = -\frac{3}{2}$, calculate the sum to infinity of the given series.

Sigma Notation (Geometric Sequence)

Ngaphambilini sifundile nge sigma notation lapho sithatha khona I phethini yethu siyibhale ngendlela elula , okwenza kubelula ukuthola isamba sazozonke izinombolo kwi phethini yethu(S_n) kanti iphinde inciphise ukubhaleka kwezinombolo ze phethini ngobuningi. Akukho okushintshile kwi geometric sequence kepha ukuze sikwazi ukuthola isamba sazozonke izinombolo zethu uma sibuka I sigma notation sisebenzisa I fomula yethu ye geometric series yokuthola isamba. $\left(S_n = \frac{a(r^n - 1)}{r - 1} \quad \text{noma} \quad S_n = \frac{a(1 - r^n)}{1 - r} \right)$

***Okubalulekile**

Njengoba I sigma notation yethu ibhala I phethini yethu ngendlela elula/ emfishane . Kubalulekile ukuqale I sigma uyehlukanise uma ingahlukanisiwe uveze izinombolo zephethini khona kuzobonakala ukuba I phethini yethu iyi <u>Arithmetic</u> noma iyi <u>Geometric</u> ,Lokhu kuzokwenza kubelula ukusebenzisa amafomula afanele kwi phethini efanele. Uma I phethini iyi geometric uzosebenzisa ama fomula afanele e geometric kanti uma I phethini iyi arithmetic uzosebenzisa ama fomula afanele e arithmetic futhi.

i.e $\sum_{k=1}^{5} 3^k$

$3^1 + 3^2 + 3^3 + 3^4 + 3^5$

$3 + 9 + 27 + 81 + 243$

$d = T_3 - T_2 = T_2 - T_1$

$d = 27 - 9 = 9 - 3$

$d = 18 = 6$

$\frac{27}{9} = \frac{9}{3} = r$

$r = 3 = 3$(le sigma notation eye geometric series ngoba ibanga izinombolo ezishiyana ngalo liyalingana)

(**Akuyona I arithmetic sequence ngoba ibanga izinombolo ezishiyana ngalo alilingani ngokuqhubeka kwe phethini**)

IZIBONELO :

Evaluate : $\sum_{k=1}^{20} 3^{k-2}$

$3^{1-2} + 3^{2-2} + 3^{3-2}$

$\frac{1}{3} + 1 + 3$

$\frac{T_3}{T_2} = \frac{T_2}{T_1} = r$

$\frac{3}{1} = 1 \div \frac{1}{3} = r$

$r = 3 = 3$

$S_n = \frac{a(r^n - 1)}{r - 1}$

$S_{20} = \frac{[(⅓)](3^{20} - 1)}{3 - 1}$

$S_{20} = 581130733,3$

<u>Okubalulekile</u>

Ukusebenzisa ama formula afanele kwi sigma efanele yikho okuyohlezi kusinika umsebenzi oliqiniso , ukuhlolisisa I sigma yethu yikho okubalulekile.

Determine the value of n if :

$$\sum_{r=1}^{n} 5.2^{1-r} = \frac{630}{64}$$

$$S_n = \frac{630}{64}$$

$$\sum_{r=1}^{n} 5.2^{1-r} = 5 + \frac{5}{2} + \frac{5}{4} + \ldots$$

$$S_n = \frac{a(1 - r^n)}{1 - r}$$

$$\frac{5[1 - (\tfrac{1}{2})^n]}{1 - (\tfrac{1}{2})} = \frac{630}{64}$$

$$5[1 - (\tfrac{1}{2})^n] = \frac{315}{64}$$

$$1 - (\tfrac{1}{2})^n = \frac{63}{64}$$

$$\frac{1}{64} = (\tfrac{1}{2})^n$$

$$(\tfrac{1}{2})^6 = (\tfrac{1}{2})^n$$

$$6 = n$$

UMSEBENZI WASEKHAYA

1)Determine the value of T if $T = \sum_{k=1}^{13} 3^{k-5}$

2)What is the value of S for which $\sum_{k=1}^{S} 5(3)^{k-1} = 65$?

3)Find the value of G if : $\sum_{k=1}^{G} 3(2)^{k-1} = 93$?

4)Given : $\sum_{k=1}^{n} 3(2)^{1-k} = 5{,}8125$ Calculate the value of n.

5)Evaluate : $\sum_{n=1}^{10} (\frac{1}{4})(-2)^{n-1}$

6)Evaluate : $\sum_{r=1}^{10} (2.4^{(r)})$

7) Evaluate : $\sum_{r=3}^{10} (3^{r-1})$

8)if $\sum_{k=1}^{20} 3(2)^{1-k} = p$,write down $\sum_{k=1}^{20} 24(2)^{-k}$ in terms of p.

Sigma Notation(Infinity Geometric Series)

Ngaphambilini sifundile ngephethini yethu engapheliyo eqhubeka njalonjalo ebizwa nge infinity pattern eyenzeka kwi Geometric Series, Le phethini nayo siyakwazi ukuyibhala ngendlela elula lapho ibhalwa ngaloluhlobo :

Kusho ukuthi iphethini yethu ayinalo inani Eligcina kulo iqhubeka njalo njalo.

$$\sum_{k=1}^{\infty} Tn$$

I Fomula ye phethini yethu

Kumele isamba sezinombolo ze phethini (Sum)

Inani lesikhundla sokuqala kwi phethini yethu(1st position)

*Inani= value

*Isikhundla=position

Uma sinikezwe I sigma notation ye phethini yethu engapheliyo siyakwazi ukubalela isamba sayo sisebenzisa I fomula yesamba se infinity geometric sequence : $S_{\infty} = \dfrac{a}{1 - r}$

Izibonelo ezilandelayo yizo ezizokunikezwa ulwazi olugcwele ngolesi sihloko.

IZIBONELO :

Calculate S_∞ if $\sum_{p=1}^{\infty} 8(4)^{1-p}$

$8(4)^{1-1} + 8(4)^{1-2} + 8(4)^{1-3} + \ldots$

$8 + 2 + \frac{1}{2} + \ldots$

$T_3 \div T_2 = T_2 \div T_1 = r$

$(\frac{1}{2}) \div 2 = 2 \div 8$

$(\frac{1}{2}) \times (\frac{1}{2}) = 2 \times (\frac{1}{8}) = r$

$(\frac{1}{4}) = (\frac{1}{4}) = r$

$$S_\infty = \frac{a}{1 - r}$$

$$S_\infty = \frac{8}{1 - (\frac{1}{4})}$$

$S_\infty = 8 \div (\frac{3}{4})$

$S_\infty = 8 \times (\frac{4}{3})$

$S_\infty = (\frac{32}{3})$ noma $10(\frac{2}{3})$

I sum to infinity yethu yenziwa I <u>nombolo yokuqala kwi phethini</u> yethu <u>nebanga izinombolo ezishiyana ngalo kwiphethini</u> , ngakho ke kubalulekile okuthi ube nazo zombili lezinto ukuze ukwazi ukubalela isamba sethu. Ingakho kumele uyehlukanise I sigma uthole izinombolo zephethini khona uzokwazi ukubalela isamba.

If $\sum_{k=p}^{\infty} 4.3^{2-k} = \frac{2}{9}$ and $r = (1/3)$,determine the value of p.

$a = 4.3^{2-p}$

$r = (1/3)$

$S_\infty = \frac{2}{9}$

$S_\infty = \frac{a}{1 - r}$

$\frac{4.3^{2-p}}{1 - (1/3)} = \frac{2}{9}$

$4.3^{2-p} = \frac{2}{9} \times \frac{2}{3}$

$4.3^{2-p} = \frac{4}{27}$

$3^{2-p} = (\cancel{4}/27) \times (1/\cancel{4}) = (1/27)$

$3^{2-p} = 3^{-3}$

$2 - p = -3$

$2+3 = p$

$5 = p$

UMSEBENZI WASEKHAYA

1) Given : $\sum_{k=1}^{\infty} 4(0,2)^{k-1}$

a)Write down the first three terms of the series.

b)Calculate the sum to infinity of the series.

2) Determine : $\sum_{k=1}^{\infty} 12\left(\frac{1}{5}\right)^{k-1}$

3) Consider the infinite geometric series: $45 + 40,5 + 36,45 + \cdots$

a) Calculate the value of the TWELFTH term of the series (correct to TWO decimal places).

b) Explain why this series converges.

c) Calculate the sum to infinity of the series.

d)What is the smallest value of n for which $s\infty - Sn < 1$?

4)Prove that $\sum_{k=1}^{\infty} 4.3^{2-k}$ is a convergent geometric series. Show ALL your calculations.

5)Find p :

$$\sum_{k=1}^{\infty} 27p^{k} = \sum_{t=1}^{12}(24 - 3t)$$

Quadratic Pattern

I – Quadratic Pattern uhlobo lwe phethini olwehlukaniseke ngezigaba ezintathu . Lapho lufike lube ne phethini eyi quadratic phezulu , lilandelwe yi First Difference mase kugcina I Second Difference. Ukuthola izinombolo ezilandelayo kwi phethini yethu eyi quadratic kulele ekutheni sithole izinombolo ezikwi first difference nokuthi sazi ukuba I second difference iyiyiphi inombolo.(I SECOND DIFFERENCE IBA INOMBOLO EYODWA **ENGASHINTSHI/CONSTANT** KULEYO NALEYO PHETHINI). I phethini yethu iba yilolu hlobo:

T_1 ; T_2 ; T_3 ; T_4 ; T_5 ; T_6 ; … $Tn = an^2 + bn + c$ (Quadratic)

$(T_2 – T_1)$ $(T_4 – T_3)$ $(T_6 – T_5)$ ///$Tn = a + d(n-1)$ First difference(d_1)

$(T_4 – T_3) - (T_2 – T_1)$ $(T_6 – T_5) - (T_4 – T_3)$ Second difference(d_2)

Umbuzo : **Uyibona kanjani I phethini ukuthi iyi Quadratic Pattern?**

Kwama phethini adlule sike safunda ngokubalela ibanga izinombolo ze phethini ezishiyana ngalo(d OR r) futhi sike sakudalula ukuba kumele lihlezi lilingana njalo njalo ngokuqhubeka kwe phethini.Kwi Quadratic Pattern ibanga izinombolo ezishiyana ngalo alilingani ngokuqhubeka kwe phethini

Kepha uma I phethini uyihlakaza iba nenombolo efanayo njalo njalo ngokuqhubeka kwe phethini kwisigaba sesithathu(Second Difference).

Khumbula :

*Ukuze sithole I nombolo elandelayo/izinombolo ezilandelayo kwi phethini yethu($Tn = an^2 + bn + c$) kumele I phethini siyihlakaze ukuze sazi izinombolo ezikuma First difference siphinde sazi nenombolo yethu engashintshi ekwi Second difference.(I phethini uyihlakaza ngalendlela eyenziwe ngenhla).

IZIBONELO :

1)Given the quadratic sequence : 321 ; 290 ; 261 ; 234 ; …

a) Write down the values of the next two terms of the sequence.

321 ; 290 ; 261 ; 234 ; … (209) (186)

(290 – 321) (261 – 290) (234 – 261)

(-31) **(-29)** **(- 27)** **(-25)** **(-23)**

(-29) - (-31) (-27) - (-29) **[(-27)+(2)]** **[(-25)+(2)]**

(2) **(2)** **(2)** **(2)**

NB

*Ukuthola izinombolo ezilandelayo kwiphethini I quadratic kulele ekutheni wazi izinombolo ezikwi first difference kanye nenombolo yethu engashintshi eku second difference.

1)Ukuthola I nombolo elandelayo kwi first difference uma I second difference seyaziwa sike sithathe I nombolo egcinile kwi first difference siyihlanganise ne second difference ukuthola I nombolo engaphambili/elandelayo.

2)Ukuthola I nombolo elandelayo kwi quadratic uma sesiyazi I first difference elandelayo sike sithathe inombolo egcinile kwi quadratic siyihlanganise ne first difference ukuthola inombolo elandelayo.

<u>Calculating the n^{th} Term of the QUADRATIC SEQUENCE</u>

Kulesi sihloko yilapho sisuke sesigxile ekutheni sithole I fomula/General Term yaleyo naleyo quadratic pattern($T_n = an^2 + bn + c$) , kepha ukuthola kwethu I fomula kulele ekutheni sikwazi ukwehlukanisa I phethini yethu ngezigaba ezintathu(Quadratic , 1^{st} difference , 2^{nd} difference) . Njengoba I fomula yethu ye quadratic inezinombolo ezingaziwa oku a , b kanye no c . Kulesi sihloko sisuke sigxile ekutholeni zona.

Lapho sizithola kanje :

1) $2a = 2^{nd}$ difference

2) $3a + b = T_1$ (1^{st} difference)

3) $a + b + c = T_1$ (Quadratic)

<u>NB</u>

*Kubalulekile ukuqala ngokubalela u <u>a</u> ngoba nguwo odingakalayo ukuze ukwazi ukubale u <u>b</u>.

*Kubalulekile ukulandela ngokubalela u <u>b</u> ngoba yiwo no <u>a</u> abadingakalayo ukuze ukwazi ukubalela u <u>c</u>.

*Uma sekufeziwe ukubalela lezi zinombolo kubalulekile ukuba zifakwe ngobunjalo bazo kwi fomula yethu($T_n = an^2 + bn + c$).

i.e

1)Given the quadratic sequence : 321 ; 290 ; 261 ; 234 ; …

I) Determine the general term of the sequence in the form $Tn = an^2 + bn + c$.

321 ; 290 ; 261 ; 234 ; … (209) (186)

(290 – 321) (261 – 290) (234 – 261)

(-31) **(-29)** **(- 27)** **(-25)** **(-23)**

(-29) - (-31) (-27) - (-29) [**(-27)+(2)**] [**(-25)+(2)**]

(2) **(2)** **(2)** **(2)**

1) 2a = 2^{nd} difference   2) $3a + b = T_1$(1^{st} difference)

2a = **(2)**   3(1) + b = **(-31)**

a = 1   b = - 34

3) $a + b + c = T_1$(Quadratic)

1 + (-34) + c = 321

-33 + c = 321

c = 354

$$Tn = n^2 – 34n + 354$$

II)Which term(s) of the sequence will have a value of 74?

$Tn = n^2 - 34n + 354$

$74 = n^2 - 34n + 354$

$0 = n^2 - 34n + 280$

$0 = (n - 14)(n - 20)$

$n = 14$ or $n = 20$

II)Which term in the sequence has the least value ?

$f'(n) = 0$

$2n - 34 = 0$

$2n = 34$

$n = 17$

<u>Term 17 will have the smallest value</u> .

UMSEBENZI WASEKHAYA

1) The sequence 3 ; 9 ; 17 ; 27 ; …. is a quadratic sequence.

a) Write down the next two terms.

b) Determine an expression for the n^{th} term of the sequence.

c)What is the value of the first term of the sequence that is greater than 269?

2) Given the sequence: 5 ; 12 ; 21 ; 32 ; .....

a)Determine the formula for the n^{th} term of the sequence.

b) Determine between which two consecutive terms in the sequence is the first difference equal to 245.

3)Consider the number pattern : 3 ; 13 ; 31 ; 57 ; 91 ; …

a)Determine the general term for this pattern.

b)Calculate the 12^{th} term of this pattern.

c)Which term is equal to 241?

4)Consider : 9 ; 19 ; 33 ; 51 ; …

a)Write the next Two terms of the pattern.

b)Determine the n^{th} term of the sequence.

c)Prove that all terms of the quadratic sequences are odd.

5)A quadratic pattern has a second term equal to 1 , a third term equal to -6 and a fifth term equal to -14.

a) Calculate the second difference of this quadratic pattern and determine the values of the first and fourth terms of the pattern.

b) Determine the n^{th} term of the sequence.

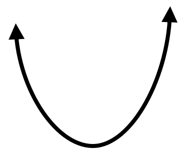
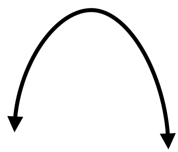
6) Consider the following quadratic sequence: 6 ; x ; 26 ; 45 ; y ; …
Calculate the values of x and y and determine the n^{th} term of the pattern.

Chapter 2

Functions

Parabola / Quadratic Function

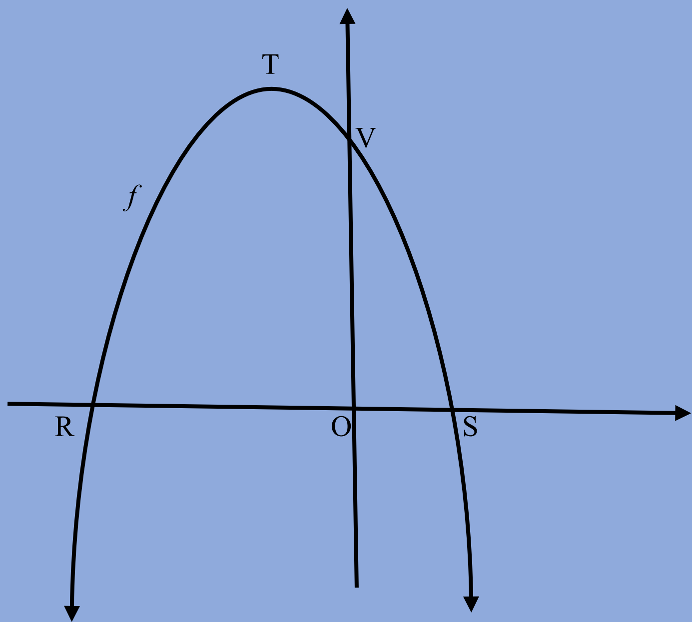
I – Quadratic / Parabola function uhlobo lwe graph enqamula ama x-axis ezindaweni ezimbili , kepha le graph I phinde ibe nendawo lapho ijika khona okuke kubizwe nge *Turning Point* lapho leli phoyinti kunezinombolo ezimbili okuyi nombolo yama x – axis nenombolo yama y- axis(x ; y). Nansi imininingwane okumele uyazi mayelana naloluhlobo lwe graph.

1	*Formulae*	$y = ax^2 + bx + c$ $y = a(x - x_1)(x - x_2)$ $y = a(x - p)^2 + q$
2	a	**Positive +a**  **Negative -a** 
3	c	y intercept value
4	p	x turning point value
5	q	y turning point value
6	*Turning point*	$x = \frac{-b}{2a}$ or (p ; q)
7	*x asymptote*	$x = p$
8	*y asymptote*	$y = q$

9	*Axis of symmetry*	$x = \frac{-b}{2a}$ or $x = p$
10	*Finding the equation of the graph*	When given the turning point $y = a(x - p)^2 + q$ When not given the turning point $y = a(x - x_1)(x - x_2)$
11	*Finding x turning point of the graph(p)*	When given x-axis $x = \frac{x_1 + x_2}{2}$
12	*Length*	$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
13	*Average gradient*	$m = \frac{y_2 - y_1}{x_2 - x_1}$
14	$x_1 ; x_2$	x intercept values
15	*m*	*Gradient*
16	*Perpendicular lines*	$m_1 \times m_2 = -1$

IZIBONELO :

The diagram below shows the graph of $f(x) = -x^2 - 2x + 8$. R and S are x – intercepts and V the y – intercept of f. T is the turning point of f.



1)Determine the length of RS.

2)Determine the coordinates of T.

3)The gradient of the tangent to the graph f at a point W is equal to 2.

3.1)Determine the coordinate of W.

3.2)Determine the equation of a straight line, g , which is perpendicular to the tangent and passing through V.

1) $f(x) = -x^2 - 2x + 8$

$0 = -x^2 - 2x + 8$

$x^2 + 2x - 8 = 0$

$(x + 4)(x - 2) = 0$

$x = -4$ or $x = 2$

R(-4;0) and S(2;0)

$RS=\sqrt{(-4-2)^2+(0-0)^2}$

RS = 6 units

2) $x = \dfrac{-4+2}{2}$

$= -1$

$f(-1) = -(-1)^2 - 2(-1) + 8$

$y = 9$

T(-1;9)

noma

$f(x) = -x^2 - 2x + 8$

$x = \dfrac{-b}{2a}$

$= \dfrac{-(-2)}{2(-1)} = -1$

$f(-1) = -(-1)^2 - 2(-1) + 8 = 9$

T(-1;9)

3.1) $f(x) = -x^2 - 2x + 8$

$f'(x) = -2x - 2$

$-2x - 2 = 2$

$x = -2$

$f(-2) = -(-2)^2 - 2(-2) + 8$

$= 8$

W(-2;8)

3.2) $g(x) = mx + c$

$m_1 \times m_2 = -1$ *(perpendicular lines)*

$2 \times m_2 = -1$

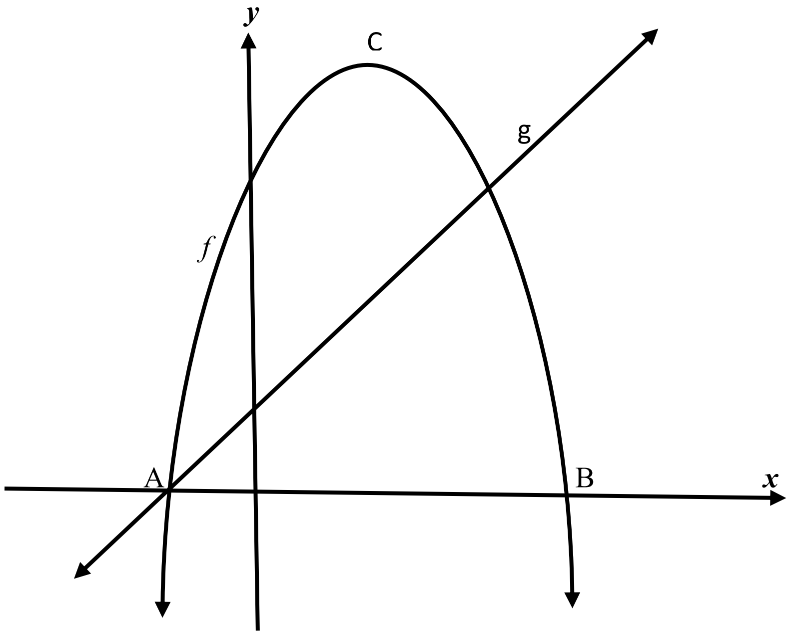
$m_2 = -\frac{1}{2}$

$c = 8$

$y = -\frac{1}{2}x + 8$

UMSEBENZI WASEKHAYA

1)Sketched below are the graph of $f(x) = -2x^2 + 4x + 16$ and $g(x) = 2x + 4$. A and B are the x – intercepts of f . C is the turning point of f.



a)Calculate the coordinates of A and B.

b)Determine the coordinates of C, the turning point of f.

c)Write down the range of f.

d)The graph of $h(x) = f(x + p) + q$ has a maximum value of 15 at $x = 2$. Determine the values of p and q.

e)Determine the equation of g^{-1} , the inverse of g , in the form y = …

f)For which value(s) of x will $g^{-1}(x).g(x) = 0$?

g)if $p(x) = f(x) + k$, determine the value(s) of k for which p and g will NOT intersect.

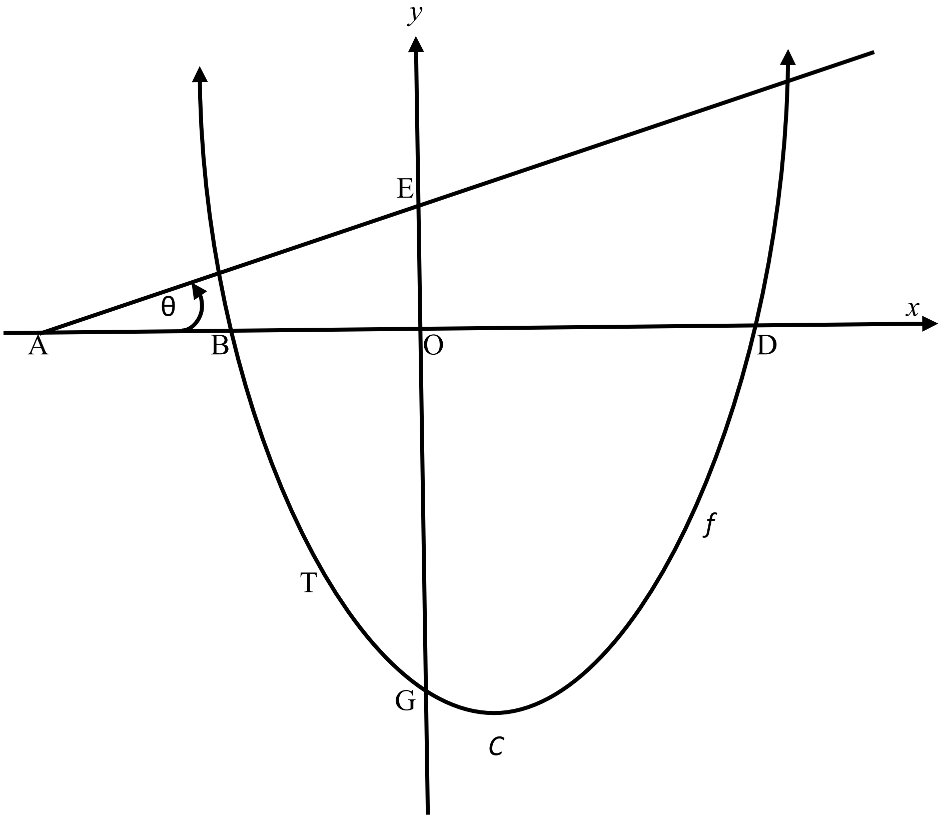
2)The graph of $f(x) = (x + 4)(x - 6)$ is drawn below.

The parabola cuts the x – axis at B and D and y – axis at G.

C is the turning point of f.

Line AE has an angle of inclination of θ and cuts the x – axis and y – axis at A and E respectively.

T is a point on f between B and G.



a)Write down the coordinates of B and D.

b)Calculate the coordinates of C.

c)Write down the range of *f*.

d)Given that $\theta = 14{,}04°$ and the tangent to *f* at T is perpendicular to AE.

e)Calculate the gradient of AE , correct to TWO decimal places.

f)Calculate the coordinates of T.

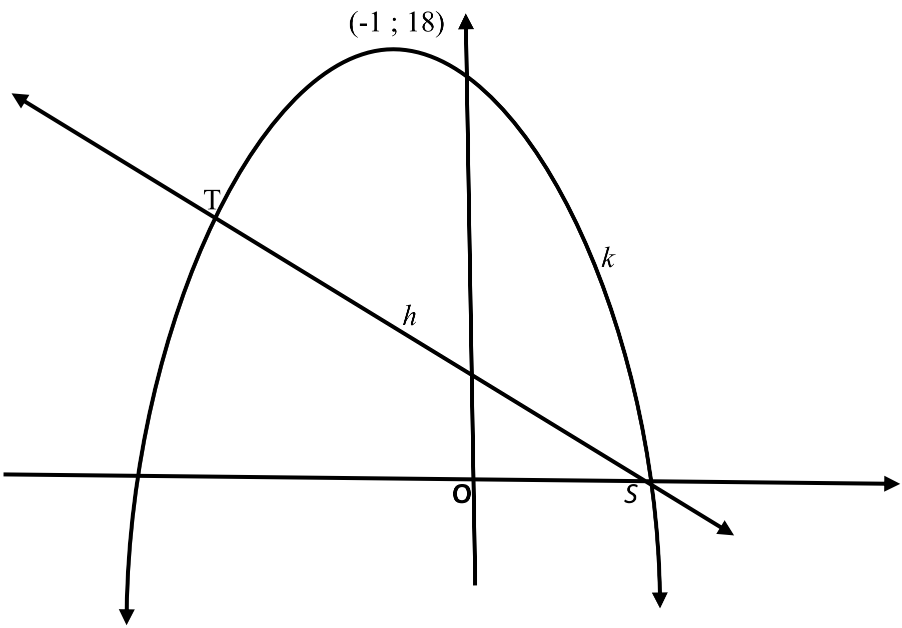
g)A straight line , g , parallel to AE , cuts *f* at K(-3 ; -9) and R.

Calculates the x – coordinates of R.

3)Sketched below are the graphs of $k(x) = ax^2 + bx + c$ and $h(x) = -2x + 4$.

Graph k has a turning point at (-1 ; 18). S is the x-intercept of h and k.

Graphs h and k also intersect at T.



a)Calculate the coordinates of S.

b)Determine the equation of *k* in the form y = a $(x + p)^2 + q$

c) if $k(x) = -2x^2 - 4x + 16$, determine the coordinates of T.

d)Determine the value(s) of x for which $k(x) < h(x)$?

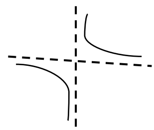
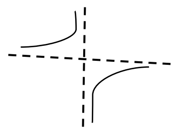
e) It is further given that *k* is the graph of $g'(x)$.

e1) For which values of x will the graph of g be concave up ?

e2)Sketch the graph of g , showing clearly the x – values of the turning points and the point of inflection.

Hyperbola Function

I hyperbola function uhlobo lwe function oluba nemidwebo/ama-graphs amabili ngasosonke iskhathi kwisiphambano sethu sokudweba imidwebo/ama graphs *(Cartesian plane)*. Indlela lemidwebo / ama-graphs ama ngayo kulele ekutheni inombolo ka **a** kwi fomula yethu iyiqiniso*(positive(+))* noma ayilona iqiniso*(negative(-))* . Nansi imininingwane okumele uyazi mayelana naloluhlobo lwe graph .

1	Formulae	$y = \frac{a}{x - p} + q$
2	*a*	**Positive *a***  **Negative *a*** 
3	*p*	*x-asymptote value*
4	*q*	*y-asymptote value*
5	*x-asymptote*	$x = p$
6	*y-asymptote*	$y = q$
7	*Axis Of Symmetry*	*Increasing function* $y = x + k$ *or* $y = (x - p) + q$

		decreasing function $y = -x + k$ *or* $y = -(x - p) + q$
8	Steps for drawing the hyperbola graph	x – intercept y – intercept x – asymptote y - asymptote
9	Domain	$x \in R$, $x \neq p$ or $(-\infty ; p) \cup (p;\infty)$ or $x < p$ or $x > p$
10	Range	$y \in R$, $y \neq q$ or $(-\infty ; q) \cup (q;\infty)$ or $y < q$ or $y > q$

ISIBONELO :

Given : $f(x) = \dfrac{-1}{x-3} + 2$

1) Write down the equation of the asymptote of f.

2) Write down the domain of f .

3) Determine the coordinates of the x-intercept of f.

4) Determine the coordinates of the y-intercept of f.

5) Draw the graph of f. Clearly show ALL the asymptotes and intercepts with the axes.

1) $x - 3 = 0 \quad x = 3$

$y = 2$

2) $x \in R$, $x \neq 3$

Or

$x \in (-\infty ; 3) \cup (3 ; \infty)$

or

$x < 3$ or $x > 3$

3) $0 = \dfrac{-1}{x-3} + 2$

$-2x + 6 = -1$

$x = (7/2)$

x – int : $(7/2 ; 0)$

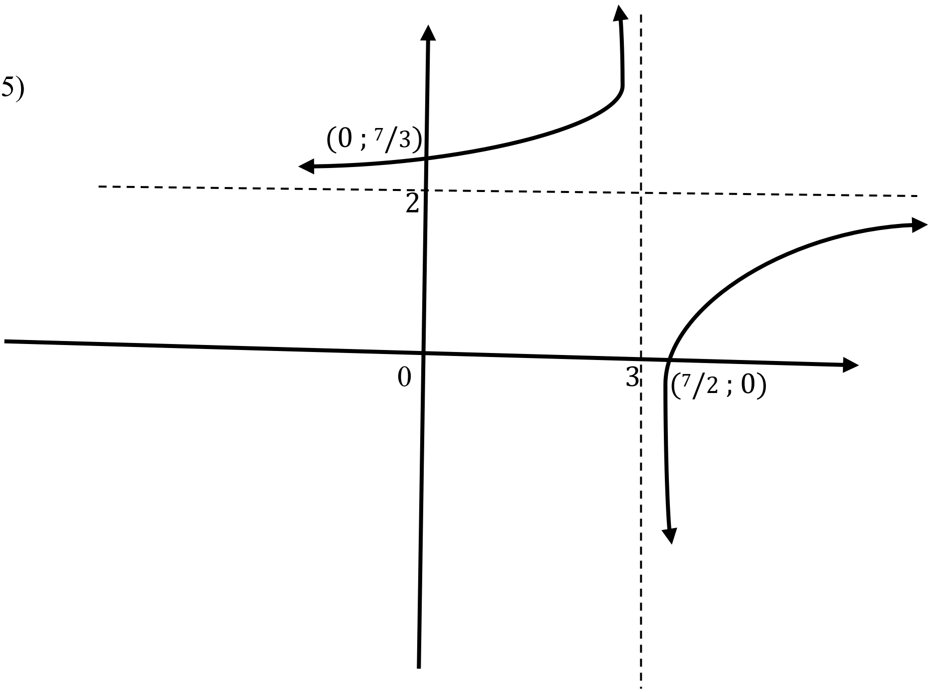
4) $y = \dfrac{-1}{0-3} + 2$

$y = (1/3) + 2$

$y = (7/3)$

y int : $(0 ; 7/3)$

5)



UMSEBENZI WASEKHAYA

1) Given : $f(x) = \dfrac{2}{x+1} - 3$

1.1) Calculate the coordinates of the y – intercept of f.
1.2) Calculate the coordinates of the x – intercept of f.
1.3) Sketch the graph of f . Clearly showing the asymptotes and the intercepts with the axes.
1.4) One of the axes of symmetry of f is a decreasing function. Write down the equation of the axis of symmetry.

2) Consider the function $f(x) = \dfrac{1}{x} - 2$

2.1) Determine :

a) The equation of the asymptotes

b) The coordinates of the x – intercepts

2.2) Sketch the graph

2.3)Write down :

a) The domain and range

b) The lines of symmetry

y = $x + c$ and y = - $x + c$

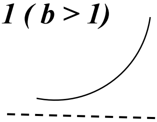
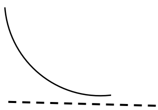
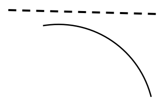
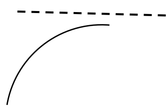
3) Consider the function $f(x) = \dfrac{-4}{x} + 1$

3.1) Firstly determine the equation of the asymptotes , coordinates of the x – intercepts and then sketch the graph of f .

3.2) If the graph of f is reflected by the line having the equation y = - $x + c$, The new graph coincides with the graph of $f(x)$. Determine the value of c.

EXPONENTIAL FUNCTION

Isifundo sama exponents isifundo ebesifundwa ngokuphindelela kulamabanga adlulile , I – Exponential function I - Function ebhekise kuso isifundo sama exponents . Lapho sikhanda I graph sisebenzise uhlobo lwe exponent esinikezwe lona. Nansi imininingwane okumele uyazi mayelana naloluhlobo lwe graph.

1	*Formulae*	$y = ab^{x-p} + q$
2	a	***Positive a*** ***When b is greater than 1 (b > 1)***  **When $0 < b < 1$**  ***Negative a*** **When b > 1**  **$0 < b < 1$** 
3	q	y – asymptote value

Exponential And Logarithmic Form

I Exponential Graph uhlobo lwe graph esikhonayo ukulijija noma siliphendule ukuze sikwazi ukwenza ama Graph amabili sibe sisebenzise I Formula eyodwa vo , Ngamanye amazwi siyakwazi ukwenza I Exponential Graph ibe newele layo ngokujija nje I Formula . Loluhlobo lwesbili lwe graph esiba nalo uma sesishintshe I Formula yethu sike silibize nge INVERSE FUNCTION . Okuchaza ukuthi I Function esuke isiphenduliwe. Lokhu sikhona ukukwenza ngokusebenzisa I Concept yama Logs .

Determining the Inverse Function

GIVEN : $f(x) = b^x$

Step1 Shintshanisa u x no y : : $x = b^y$

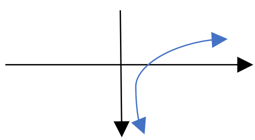
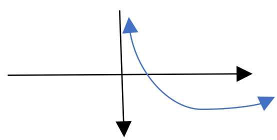
Step2 Enza u y i subject of the Formula : $y = \text{Log}_b x$

Step3 INVERSE FUNCTION Formula : $f(x)^{-1} = \text{Log}_b x$

*Final Graphs : Exponential Graph : $F(x) = b^x$

: Inverse Function : $F(x)^{-1} = \text{Log}_b x$

Graphs : $f(x)^{-1} = Log_b x$

If b > 1	If 0< b <1
	

IZIBONELO :

Given $f(x) = ({}^{1}/{}_{5})^{x}$

1.Find the equation of $f(x)^{-1}$

$f(x) = ({}^{1}/{}_{5})^{x}$

$x = ({}^{1}/{}_{5})^{y}$

$y = \mathrm{Log}_{(1/5)}x$

$f(x)^{-1} = \mathrm{Log}_{(1/5)}x$

2.State the intercept(s) for each graph.

Graph $f(x)$: (0,1)

x- Intercept : $x = 0$

y- Intercept : $f(x) = ({}^{1}/{}_{5})^{0} = 1$

Graph $f(x)^{-1}$: (1,0)

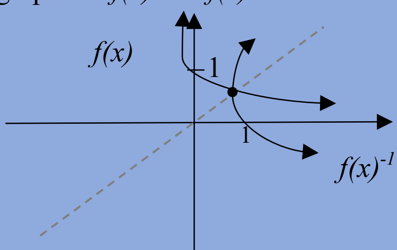
x- Intercept : $0 = \mathrm{Log}_{(1/5)}x$

$({}^{1}/{}_{5})^{0} = x$

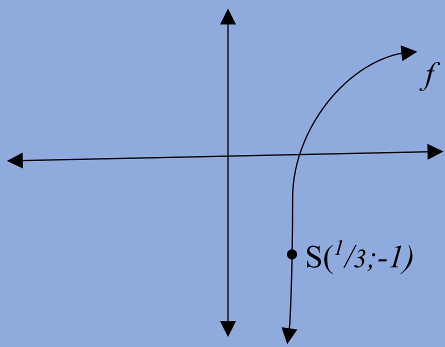
$x = 1$

$y = 0$

3)Sketch the graphs of $f(x)$ and $f(x)^{-1}$ on the same set of Axes



Given : $f(x) = \mathrm{Log}_a x$ where a > 0. S($^1/3$;-1) is a point on the graph of f.



1)*Prove that a = 3*

$f(x) = Log_a x$

$-1 = Log_a(1/3)$

$a^{-1} = 1/3$

$a^{-1}=1/3$

$a^{-1}=3^{-1}$

$a=3$

*Uma unikezwe I Phoyinti elinama coordinates kumdwebo wethu waphinde wanikezwa ne formula engakabi no **a** oyinombolo/value . Kubalulekile ukusebenzisa I Phoyinti lethu silifake kwi formula ukuze sithole inani la **a** wethu.

2) Write down the equation of h , the inverse of f , in the form of y = . . .

$f(x) = \mathrm{Log}_a x$

$x = \mathrm{Log}_3 y$

$y = 3^x$

3)If $g(x) = -f(x)$, determine the equation of g .

$g(x) = -\text{Log}_3 x$

4) Write down the domain of g .

Domain : $x > 0$

5) Determine the values of x for which $f(x) \geq -3$

$f(x) \geq -3$

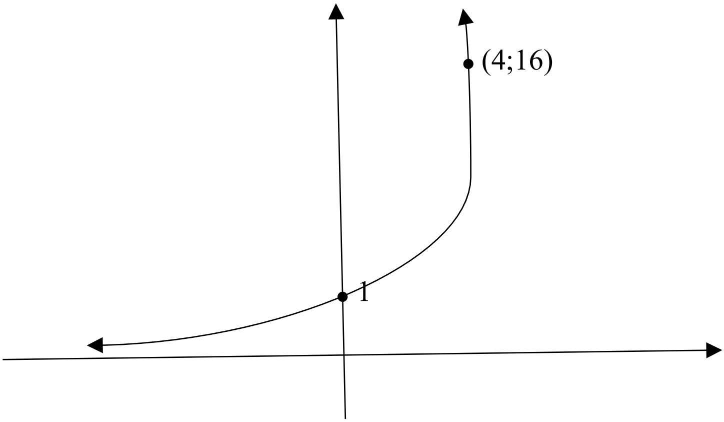
$\text{Log}_3 x \geq -3$

$x \geq 3^{-3}$

$x \geq \frac{1}{27}$

UMSEBENZI WASEKHAYA

Sketched below is the graph of $f(x) = k^x$; $k > 0$. The Point (4;16) lies on f.



1)Determine the value of k.

2)Graph g is obtained by reflecting graph f about the line $y = x$. Determine the equation of g in the form of y= . . .

3) Sketch the graph g. Indicate on your graph the coordinates of two points on g.

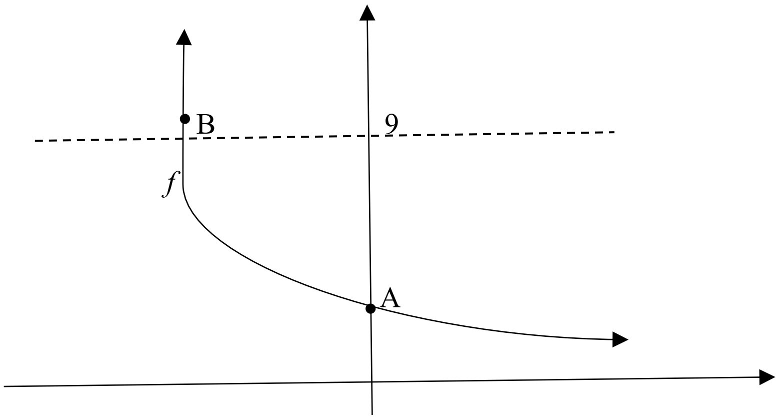
4) Use your graph to determine the value(s) of x for which :

a) $f(x) \times g(x) > 0$

b)$g(x) \leq -1$

5)If $h(x) = f(-x)$, Calculate the value of x for which $f(x) - h(x) = \frac{15}{4}$

2) The graph of $f(x) = 3^{-x}$ is sketched below . A is the y-intercept of f. B is the point of intersection of f and the line y = 9.



1)Write down the coordinates of A.

2) Determine the coordinates of B.

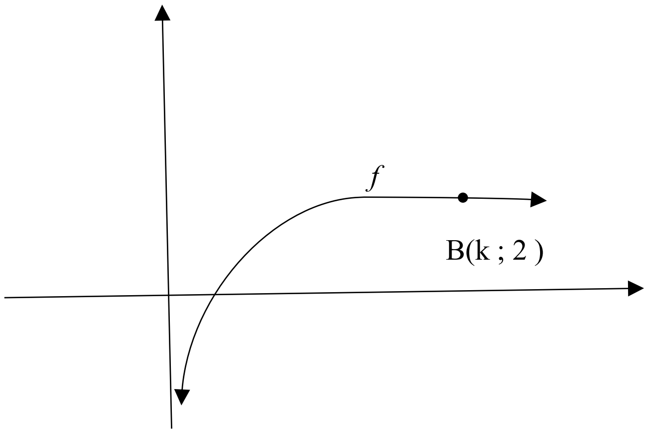
3) Write down the domain of f^{-1}

4) Describe the translation from f to $h(x) = \dfrac{27}{3^x}$

5)Determine the values of x for which $h(x) < 1$.

3) The graph of $f(x) = \text{Log}_4 x$ is drawn below.

B(k ; 2) is a point on f.



1) Calculate the value of k.

2) Determine the values of x for which $-1 \leq f(x) \leq 2$

3) Write down the equation of f^{-1}, the inverse of f, in the form of y = …

4) For which values of x will $x.f^{-1}(x) < 0$?

Chapter 3

ANALYTICAL GEOMETRY

ANALYTICAL GEO

Analytical Geometry isihloko esigxile kakhulu ekusebenziseni I algebra ukuxazulula izinkinga ezikwama shapes wethu . Lapho sisuke sisebenzisa ama formula ukuqhamuka nezimpendulo eziyiqiniso/*correct answers*. Ama formula esiwasebenzisayo yilawa angezansi :

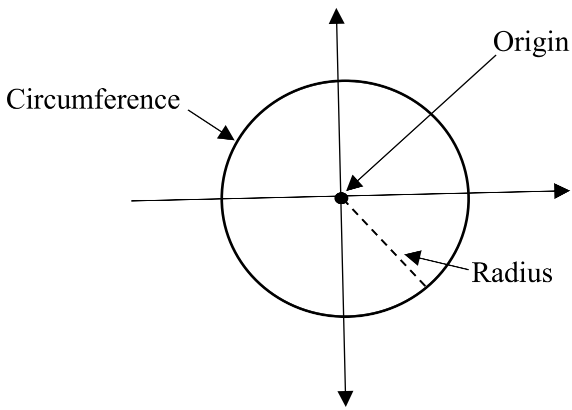
Formula Names	Equations	Examples
Distance	$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$	Given N(3;6) and K(-1;-2). Calculate their distance. $NK = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ $NK = \sqrt{(3+1)^2 + (6+2)^2}$ $NK = \sqrt{80}$
Gradient of a line	$m = \dfrac{y_2 - y_1}{x_2 - x_1}$	Calculate the gradient of NK. $m = \dfrac{y_2 - y_1}{x_2 - x_1}$ $= \dfrac{6+2}{3+1}$ $m = 2$
Midpoint	$m = \dfrac{x_1 + x_2}{2} ; \dfrac{y_1 + y_2}{2}$	Determine the coordinates of D , The Midpoint of NK. $m = \dfrac{x_1 + x_2}{2} ; \dfrac{y_1 + y_2}{2}$ $= \dfrac{3-1}{2} ; \dfrac{6-2}{2}$ $= 1 ; 2$ D(1;2)
Equation of a line	$y = mx + c$ or $y - y_1 = m(x - x_1)$	Determine the equation of line NK. $y = mx + c$ $y = 2x + c$ $6 = 2(3) + c$ $c = 0$ $y = 2x$
Parallel lines(AB//CD)	$m_{AB} = m_{CD}$	Given S(6,8) and R(4,4). Prove that NK//SR $m = \dfrac{y_2 - y_1}{x_2 - x_1}$ $= \dfrac{8-4}{6-4}$ $m = 2$ $m_{SR} = m_{NK} = 2$
Perpendicular lines(AB⊥CD)	$m_{AB} \times m_{CD} = -1$	PK is perpendicular to NK . Determine the gradient of PK $m_{NK} \times m_{PK} = -1$ $2 \times m_{PK} = -1$ $m_{pk} = -(1/2)$

Inclination	$m = tan\theta$	Line NK has an angle of inclination which is θ. Dertemine the value of θ $m = tan\theta$ $2 = tan\theta$ $\theta = 63{,}43$

Circles

Isifundo sama circles sihlukene kabili. Esokuqala yilapho sifunda khona nge circle/indilinga esuke ikwi Origin(*Lapho iphakathi nendawo lendilinga lisuke likwi number line ye Cartesian Plane)*. Kanti esesibili isifundo yilapho sisuke sesifunda nge circle/indilinga esuke ingekho kwi Origin*(Iphakathi nendawo lendilinga lisuke lingekho kwi number line ye Cartesian Plane)*.

<u>Circle with centre Origin</u>



Ukubekwa kwe Circle/indlilinga kwi Origin kuchaza ukuthi leyondilinga ayinayo I x-value kanye ne y-value . Lokho kuchaza ukuthi ama coordinates ephakathu nendawo le circle yethu awu (0;0). Sisebenzisa le formula engezansi ukubalela indilinga esuke ikwi Origin :

$$x^2 + y^2 = r^2$$

IZIBONELO :

1)The circle with the centre 0 has point M(5;2) on its circumference . Determine the equation of a circle.

$x^2 + y^2 = r^2$

$(5)^2 + (2)^2 = r^2$

$25 + 4 = r^2$

$29 = r^2$

Equation : $x^2 + y^2 = 29$

2) Determine the equation of a circle with the centre 0 and passing through Point (-3 ; 4)

$x^2 + y^2 = r^2$

$(-3)^2 + (4)^2 = r^2$

$9 + 16 = r^2$

$25 = r^2$

Equation : $x^2 + y^2 = 25$

3) Determine the equation of a circle with the centre 0 and passing through Point (16 ;5)

$x^2 + y^2 = r^2$

$(16)^2 + (5)^2 = r^2$

$256+ 25 = r^2$

$281 = r^2$

Equation : $x^2 + y^2 = 281$

CIRCLES NOT CENTRED AT THE ORIGIN

Ama Circles asuke engekho kwi Origin/*Number Line* kusuke kwama Circles ane x-value kanye ne y-value yephakathi nendawo (x ; y). Sisebenzisa loluhlobo lwe formula ukubalela i Circle esuke ingekho kwi Origin :

$$(x - a)^2 + (y - b)^2 = r^2$$

a = x-value

b = y-value

IZIBONELO :

1)The equation of a circle is $(x + 5)^2 + (y - 3)^2 = 25$. Determine the coordinates of the centre and the length of the radius

NB) Njengoba I equation yethu ihleliwe , izinombolo eziphakathi kuma brackets zimele ama coordinates e centre yethu. Kanti engaphandle inombolo imele I radius. Ngakho ke :

Coordinates of the Centre(-5 ; 3)

Radius : $r^2 = 25$

$r = 5$

2) Given the coordinate of the centre (8 , -6) and 9 as the radius.

Determine the equation of the circle

$(x - 8)^2 + (y + 6)^2 = (9)^2$

$(x - 8)^2 + (y + 6)^2 = 81$

COMPLETING A SQUARE

I Formula ye Circle/indilinga esuke ingekho kwi origin ayihlezi ihlanganisiwe $[(x - a)^2 + (y - b)^2 = r^2]$. Kuke kwenzeke itholakale ihlukanisiwe lapho ke kusuke sekumele uyihlanganise ibe sesimweni . Uku Complete I Square iyona ndlela yokuyihlanganisa I Formula yethu kanti lokhu sikwenza kanje :

$$x^2 + 2x + y^2 - 4y - 9 = 0$$

Step 1 : Thatha inombolo engaxubene nama alphabet uyiwelise ngaphesheya kwe equal sign.

$$x^2 + 2x + y^2 - 4y = 9$$

Step 2 : Divide nge whole number eseceleni kwa to x^2. *(kulesi isimo i whole number eseceleni kwa x^2 u 1*).

$$\frac{x^2}{1} + \frac{2x}{1} + \frac{y^2}{1} - \frac{4y}{1} = \frac{9}{1}$$

Step 3: Andisa ama calculations kwi equation yakho ngokuthatha izinombolo eziseceleni kwa x no y uzi divide ngo 2 ebese lezi zinombolo uzi multiply ngenani lazo ($\boldsymbol{x^2}$). Okwenza esandleni sokunxele kumele ukwenze nakwisandla sokudla.

$$x^2 + 2x + (2/2)^2 + y^2 - 4y + (-4/2)^2 = 9 + (2/2)^2 + (-4/2)^2$$

Step 4 : Lapha kusuke sekumele u Factorise . Izinombolo ozozithola ngaphakathi kwama brackets akho yizo ezizoba ama values akho mase u factorise.

$$(x + 1)^2 + (y - 2)^2 = 14$$

From General Form : $x^2 + 2x + y^2 - 4y = 9$

To Standard Form : $(x + 1)^2 + (y - 2)^2 = 14$

IZIBONELO :

Determine the coordinates of the centre and the length of the radius if a circle has the equation: $x^2 - 2x + y^2 + 10y + 14 = 0$.

$x^2 - 2x + y^2 + 10y + 14 = 0.$

$x^2 - 2x + y^2 + 10y = -14$

$x^2 - 2x + (-2/2)^2 + y^2 + 10y + (10/2)^2 = -14 + (-2/2)^2 + (10/2)^2$

$x^2 - 2x + (-1)^2 + y^2 + 10y + (5)^2 = -14 + 1 + 25$

$(x - 1)^2 + (y + 5)^2 = 12$

Centre : (1 , - 5)

Radius : $r^2 = 12$

$r = \sqrt{12}$

2) Write the equation of the circle in Standard Form

$x^2 + y^2 + 8x - 2y - 8 = 0$

$x^2 + 8x + y^2 - 2y = 8$

$x^2 + 8x + (8/2)^2 + y^2 - 2y + (-2/2)^2 = 8 + (8/2)^2 + (-2/2)^2$

$x^2 + 8x + (8/2)^2 + y^2 - 2y + (-2/2)^2 = 25$

$(x + 4)^2 + (y - 1)^2 = 25$

Centre : (-4 , 1)

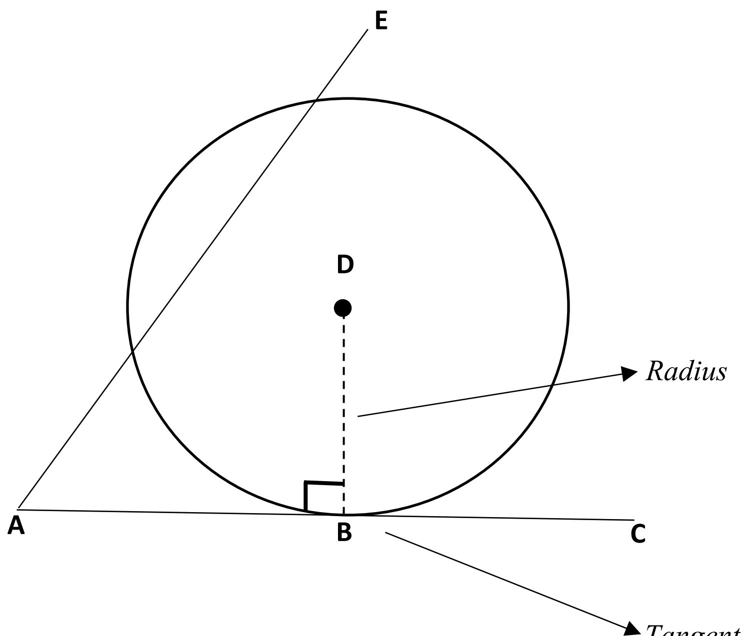
Radius : $r^2 = 25$

$r = 5$

The Equation Of A Tangent To A Circle

I tangent ulayini oqondile(*straight line)* osuke uthinta indilinga yethu/circle kwi phoyinti elilodwa . ukuhlangana kwe tangent kanye nolayini osuke usuka kwiphakathi nendawo lendilinga yethu*(Radius)* lokho kudala I angle ewu **90º** kulelo phoyinti labolayini abasuke behlangane kulo . Ukudaleka kwe angle ewu 90º kuchaza ukuthi labolayini ba perpendicular kanti sike safunda ngaphambilini ukuthi uma si multiply i gradient yolayini aba perpendicular kumele basinikeze u – *1* . Ngakho ke :

$$m_{radius} \times m_{tangent} = -1$$



$AC \perp DB$

$$m_{AC} \times m_{DB} = -1$$

Note

AE awsiyona I tangent ngoba awuyithinti I circle kwi phoyinti elilodwa . **AE** uthinta I circle kwamaphoyinti amabili.

Finding The Equation Of A Tangent

Step 1. Thola ama coordinates ephakathi nendawo lendilinga yakho (a ; b)

Step 2. Thola inani le Gradient ye Radius.

Step 3. Khumbula ukuthi i Radius i Perpendicular to tangent , Ngakho ke sebenzisa I formula $m_{radius} \times m_{tangent} = -1$ ukuthola I Gradient ye tangent njengoba usuyazi/usunayo I Gradient ye Radius.

Step 4. Thatha I Gradient ye tangent kanye nanoma yiliphi iphoyinti elikwi tangent , kokubili lokhu uku substitute kwi straight line equation ebizwa nge gradient-point $(y - y_1) = m(x - x_1)$ **.** Kumele uhlezi wenza u **y** I subject of the formula.

IZIBONELO :

Find the equation of the tangent ABC which touches a circle centre D with equation $(x - 3)^2 + (y + 1)^2 = 20$ at B(5; 3).

- Centre D(3; –1)
- $m_{radius} = \dfrac{y_2 - y_1}{x_2 - x_1} = \dfrac{3 - (-1)}{5 - 3} = 2$
- $m_{radius} \times m_{tangent} = -1$

  $2 \times m_{tangent} = -1$

  $m_{tangent} = -1/2$
- $(y - y_1) = m(x - x_1)$

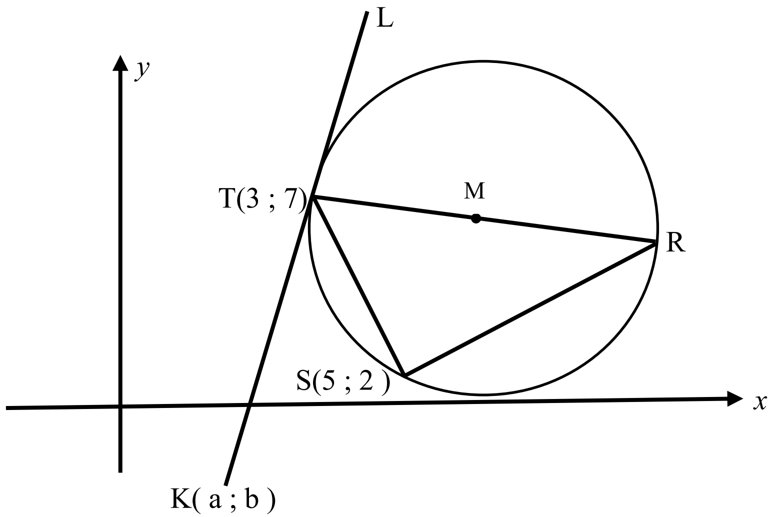
  $y - 3 = -\frac{1}{2}(x - 5)$

  $y - 3 = -\frac{1}{2}x + 2\frac{1}{2}$

  $y = -\frac{1}{2}x + 5\frac{1}{2}$

UMSEBENZI WASEKHAYA

1) In the diagram , M is the centre of the circle passing through T(3 ; 7) , R and S(5 ; 2) . RT is a diameter of the circle , K(a ; b) is a point in the 4^{th} quadrant such that KTL is a tangent to the circle at T.



1.1 Give a reason why $T\hat{S}R = 90^o$

1.2 Calculate the gradient of TS

1.3 Determine the equation of the line SR in the form of y = mx+ c .

1.4)The equation of the circle above is $(x - 9)^2 + (y - 6\frac{1}{2}) = 36\frac{1}{4}$

1.4.1) Calculate the length of TR in surd form.

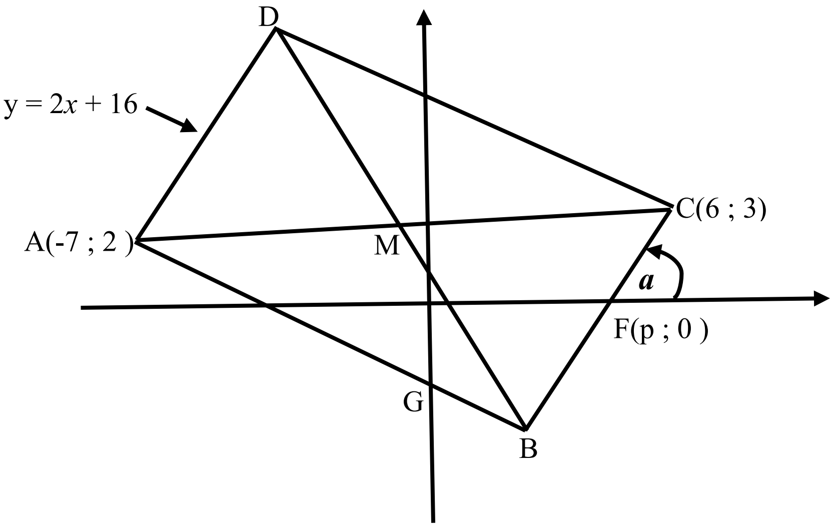
1.4.2) Calculate the coordinates of R

1.4.3) Calculate *SinR*

1.4.4) Show that b = 12a – 29

1.4.5) If TK = TR , Calculate the coordinates of K.

2) In the diagram, A(-7 ; 2), B , C(6 ; 3) and D are the vertices of rectangle ABCD. The equation of AD is $y = 2x + 16$. Line AB cuts the y-axis at G. The x-intercept of line BC is F(p ; 0) and the angle of inclination of BC with the positive x-axis is a. The diagonals of the rectangle intersect at M.



2.1 Calculate the coordinates of M.

2.2 Write down the gradient of BC in terms of p

2.3 Hence , Calculate the value of p.

2.4 Calculate the length of DB.

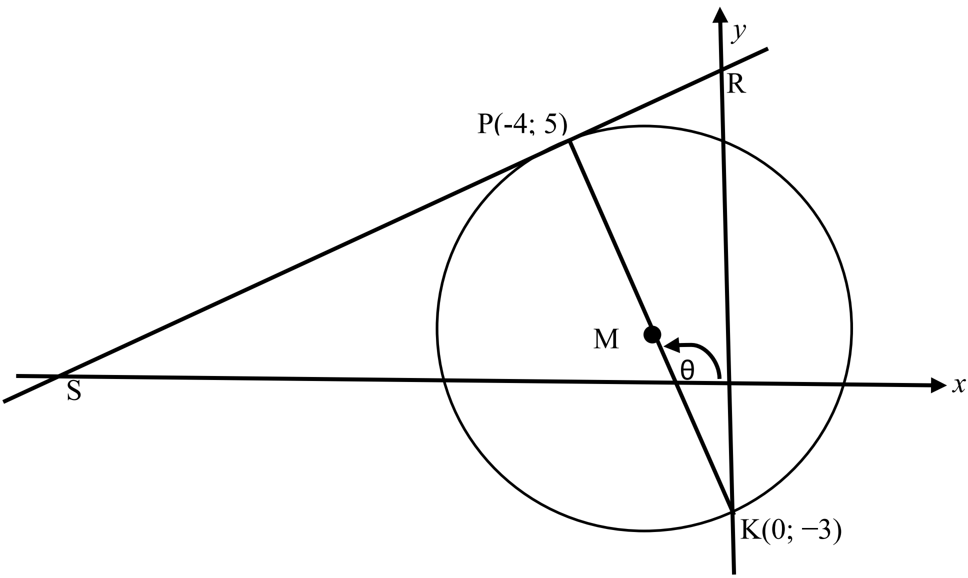
2.5 Calculate the size of a.

2.6 Calculate the size of $O\hat{G}B$.

2.7 Determine the equation of the circle passing through points D , B and C in the form $(x - a)^2 + (y - b)^2 = r^2$.

2.8 If AD is shifted so that ABCD becomes a square , will BC be a tangent to the circle passing through points A , M and B , where M is now the intersection of the diagonals of the square ABCD ? Motivate your answer.

3) In the diagram, P(-4; 5) and K(0; −3) are the end points of the diameter of a circle with centre M. S and R are respectively the x- and y-intercept of the tangent to the circle at P. θ is the inclination of PK with the positive x-axis.



3.1 Determine :

3.1.1 The gradient of SR

3.1.2 The equation of SR in the form $y = mx + c$

3.1.3 The equation of the circle in the form $(x - a)^2 + (y - b)^2 = r^2$

3.1.4 The size of PKR

3.1.5 The equation of the tangent to the circle at K in the form of y = mx + c

3.2 Determine the values of t such that the line $y = \frac{1}{2}x + t$ cuts the circle at two different points.

3.3 Calculate the area of $\triangle$ SMK

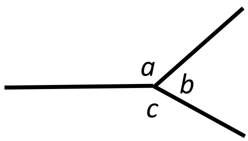
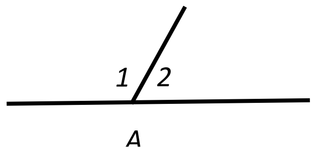
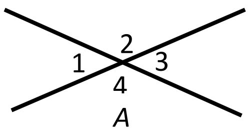
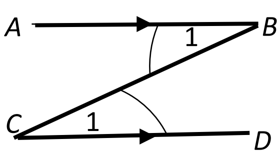
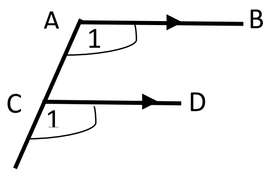
Chapter 4

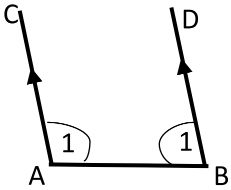
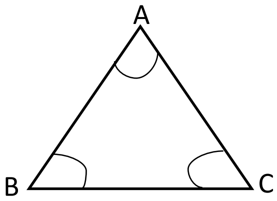
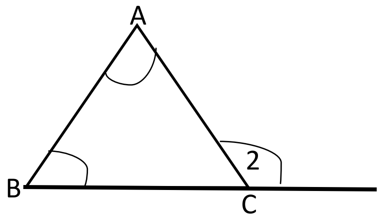
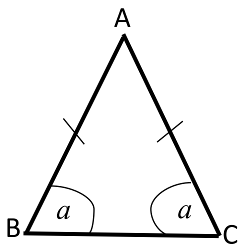
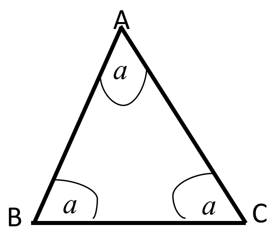
EUCLIDEAN GEOMETRY

EUCLIDEAN GEOMETRY

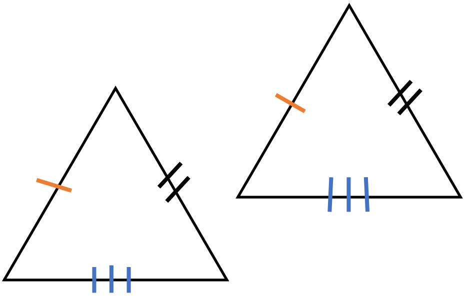
I Euclidean Geometry isifundo esasungulwa u EUCLID ongusolwazi lwezibalo wase Greek . Lesisifundo sibhekise ekusebenziseni imithetho ka Euclid ukuxazulula izinkinga ezikuma shapes wethu , Lapho sizobe sifunda ngama theorems kanye nendlela esingawasebenzisa ngayo ama theorems wethu ukuqhamuka nezixazululo . Imithetho yokuqala ye Euclidean Geometry

Ithi :

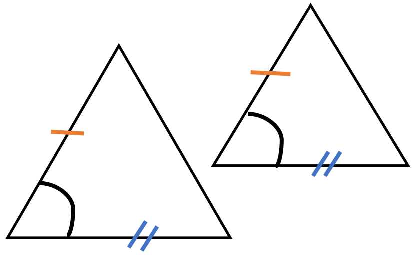
Straight Lines	
The sum of Angles around a point is 360^o  $a + b + c = 360^o$ **(<s around a point)**	Adjacent angles on a straight line are supplementary *(add to 180^o)*  $\hat{A}_1 + \hat{A}_2 = 180^o$ **(<s on straight line)**
Vertical opposite angles are equal  $\hat{A}_1 = \hat{A}_3$ and $\hat{A}_2 = \hat{A}_4$ **(Vert opp <s =)**	
Parallel Lines	
 Alternate angles are equal (**Z/N** – Shape) . If AB//CD then alternate angles are equal ($B_1 = C_1$)**(Alt <s=)**	 Corresponding angles are equal(**F-** Shape) . If AB//CD then corresponding angles are equal($\hat{A}_1 = \hat{C}_1$) (**Corresp <s;AB//CD)**

 Co-interior angles are equal (**U** – Shape) . If AC//BD then Co-interior angles are equal ($A_1 = B_1$)**(Co-int < s = ; AC//BD)**	
Triangles	
 Interior angles of a triangle are supplementary *(add to 180º)* $\hat{A} + B + \hat{C} = 180^o$ **(Int <s △)**	 Exterior angle of a triangle is equal to the sum of the interior opposite angles $C_2 = \hat{A} + \dot{B}$ **(Ext <s △)**
 The angles opposite the equal sides in a triangle are equal $\dot{B} = \dot{C}$ **(<s opp = sides)**	 If all angles in a triangle are equal , Then all the sides are equal .Also if all the sides are equal , all angles are equal If $\hat{A} = \dot{B} = \hat{C} = a$, Then AB = AC = BC

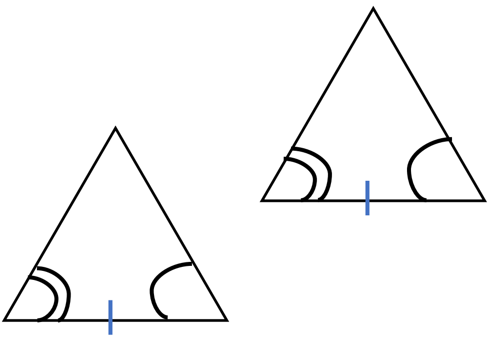
Congruency Of Triangles



If all the three sides of one triangle are equivalent to the corresponding three sides of the second triangle, then those two triangles are congruent by **SSS** rule.

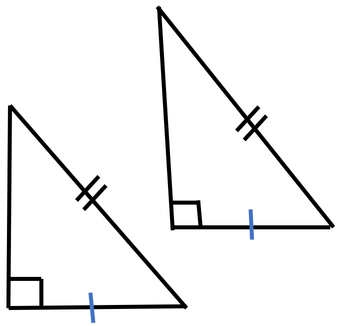


If two sides and the included angle of one triangle equals to the two sides and the included angle of another triangle . Those two triangles are congruent by **SAS** rule.

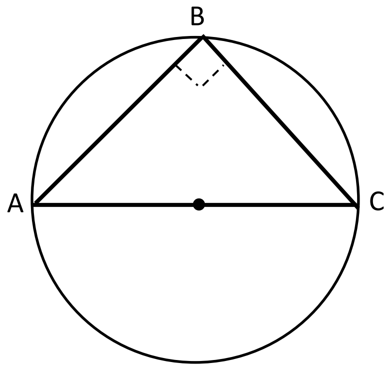
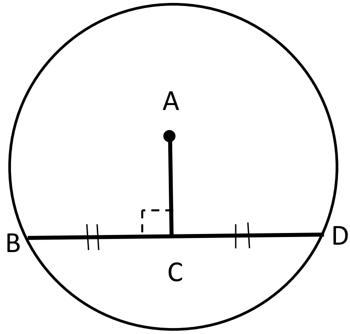


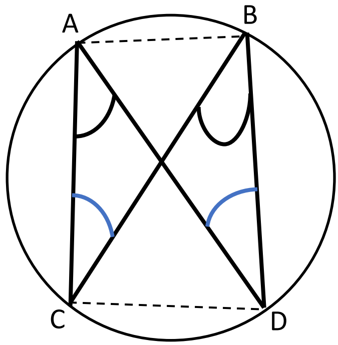
If two angles and one side of a triangle equals to two angles and one side of another triangle ,

Those triangles are congruent by **AAS** rule.



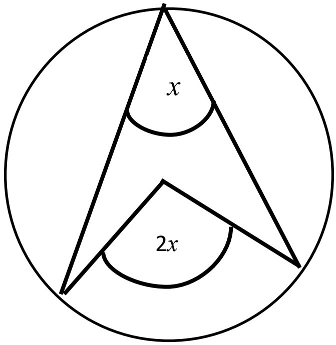
If the hypotenuse and a side of one triangles equals to the hypotenuse and a side of another triangle . Those Triangles are Congruent by **RHS** rule.

Circle Theorems	
 The angle subtended by the diameter at the circumference of the circle is 90^{o} $A\dot{B}C = 90^{O}$ **($<_{s}$ in Semi-Circle OR < subt by diam)**	 The line drawn from the centre of a circle perpendicular to a chord bisects the chord. BC = CD **(line from center ⊥ Chord)** **Also** (The line drawn from the centre of a circle to the midpoint of a chord is perpendicular to the chord)



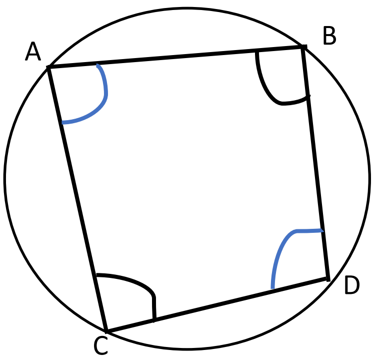
Angles subtended by a chord of the circle, on the same side of the chord are equal .

Therefore Ȁ = Ḃ And Ĉ = Ḋ **($<_s$ in the same seg.)**

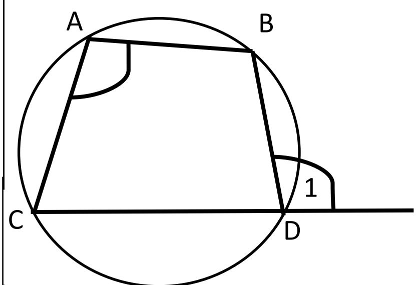


angle subtended by an arc at the centre is double the angle subtended by it at any point on the remaining part of the circle

(< at centre = 2 x < at circumference)

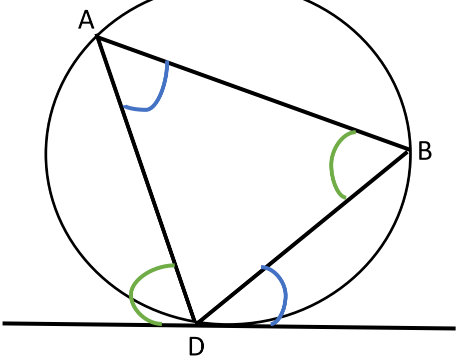


The opposite angles of a cyclic quadrilateral are supplementary *(add to 180)* . Therefore Ȁ + Ḋ = 180° and Ḃ+ Ĉ= 180° **(opp <s of cyclic quad)**



The exterior angle of a cyclic quadrilateral is equal to the interior opposite angle.

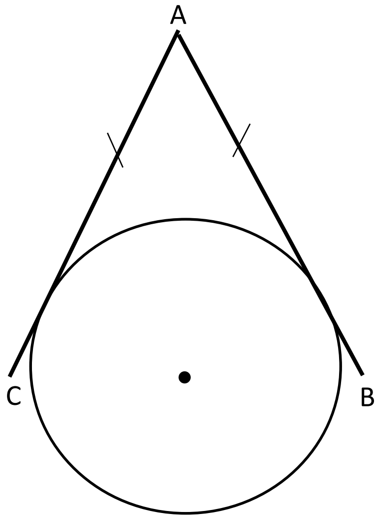
$\dot{D}_1$ = Ȁ **(ext < cyclic quad)**



The angle between the tangent to a circle and the chord drawn from the point of contact is equal to the angle in the alternate segment.

$\dot{D}_1 = \dot{B}$ **(Tan Chord Theorem)**

$\dot{D}_3 = \hat{A}$ **(Tan Chord Theorem)**



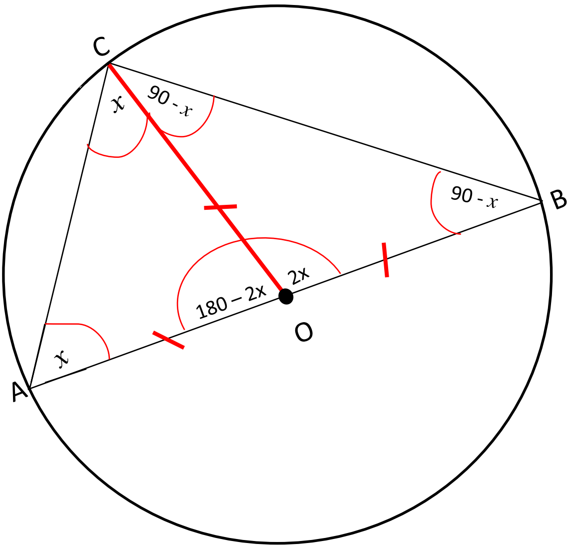
Two tangents drawn to a circle from the same point outside the circle are equal in length.

AB = AC **(Tans from same point)**

Circle Theorem Proofs

AB is the diameter of the circle . Prove that angle ACB = 90°

Construction :



OA = OB = OC (All Radii)

Let angle OAC = x

Angle ACO = x (<s opp = sides)

Angle COA = 180 – 2x (**Sum <s △**)

Angle COB = 2x (<s on straight line)

Angle OBC = $\frac{180 - 2x}{2} = 90 - x$

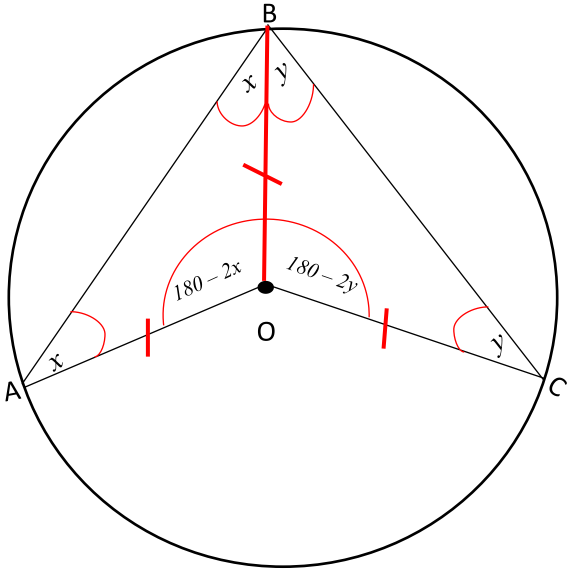
Angle OCB = 90 – x (base angles in isosceles triangle are equal)

Angle ACB = $x + 90 - x$

ACB = 90°

Prove that angle AOC = 2 × Angle ABC

Construction :



OA = OB = OC (Radii)

Let angle OAB = x

angle OBA = x (<s opp = sides)

Let angle OCB = y

angle OBC = y (<s opp = sides)

angle BOA = 180 – 2x (**Sum <s △**)

angle BOC = 180 – 2y (**Sum <s △**)

angle AOC = 360 – (180 – 2x) – (180 – 2y) (angles around point add to 360°)

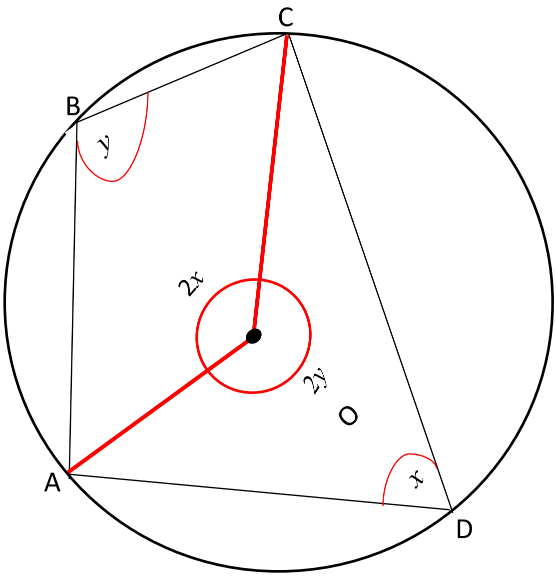
= 360 – 180 + 2x – 180 + 2y

= 2x + 2y

= 2 (x + y) = 2 × Angle ABC

Prove that Angle CDA + Angle ABC = 180º

Construction :



Let angle CDA = x

Angle COA = 2x (< at centre = 2 × < at circumference)

Let angle ABC = y

Angle COA = 2y (< at centre = 2 × < at circumference)

Ộ = [2x + 2y = 360] Angles around a point add 360º

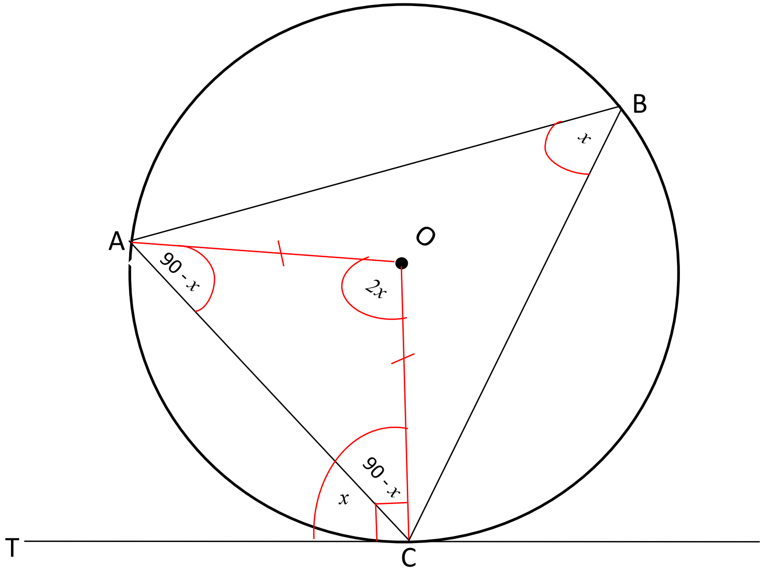
2x + 2y = 360

x + y = 180

angle CDA + angle ABC = 180º

Prove that angle ACT = angle ABC

Construction :



Let angle ACT = x

angle OCT = 90° (Radius ⊥ Tan)

angle ACT + angle ACO = Angle OCT

x + angle ACO = 90^0

angle ACO = $90^\circ - x$

angle CAO = $90^\circ - x$ (<s opp = sides)

angle AOC = $180 - (90 - x) - (90 - x)$ (**Sum <s** △)

$= 180 - 90 + x - 90 + x$

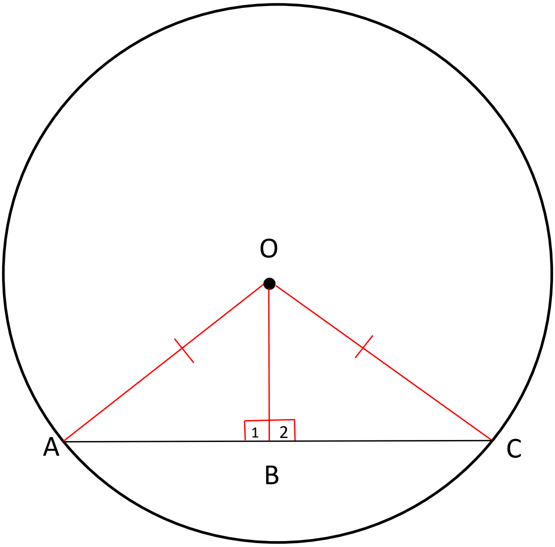
$= 2x$

Angle ABC = x (< at centre = 2 × < at circumference)

Angle ACT = Angle ABC = x

OB is Perpendicular to AC . Prove that AB = BC

Construction :



In △ OAB and △ OBC

OA = OC (Radii)

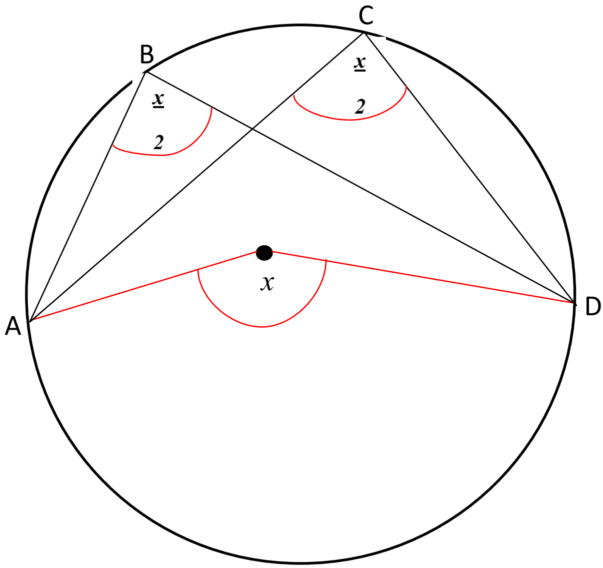
OB = OB (Common Side)

$B_1 = B_2 = 90^o$ (Given)

△ OAB ≡ △OBC (RHS)

Prove that angle ABD = angle ACD

Construction :



Let angle AOD = x

Angle ABD = $\frac{x}{2}$ (< at centre = 2 × < at circumference)

Angle ACD = $\frac{x}{2}$ (< at centre = 2 × < at circumference)

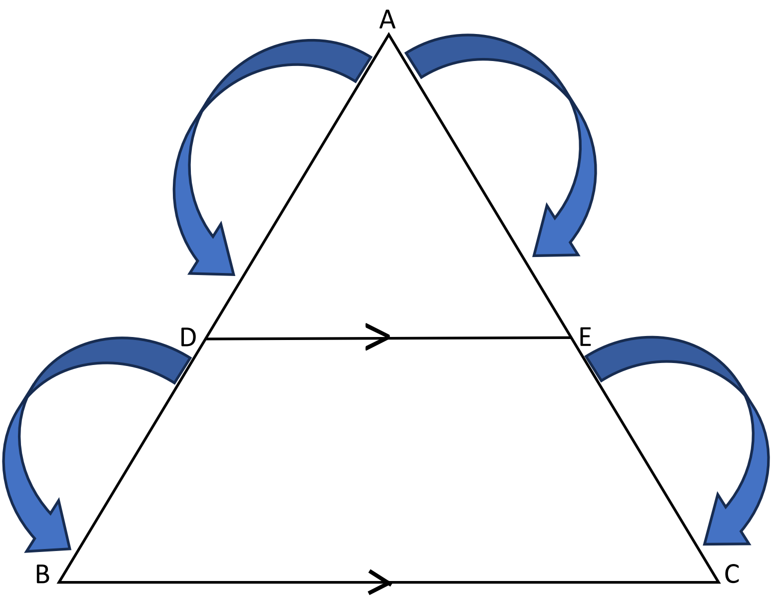
Angle ABD = Angle ACD = $\frac{x}{2}$

RATIO AND PROPORTION

PROPORTIONALITY THEOREM

I Proportionality Theorem ingenye yama theorems asetshenziswa kuma triangles kanti ichaza ukuthi : Uma kunolayini odwetshwe waxhuma amasayidi amabili e triangle wase uba parallel kwi sayidi elilodwa , Lo layini uhlukanisa I triangle wenze kube nama sayidi alinganayo ngama Ratios.

Theorem : **A line drawn parallel to one side of a triangle divides the other two sides proportionally.**



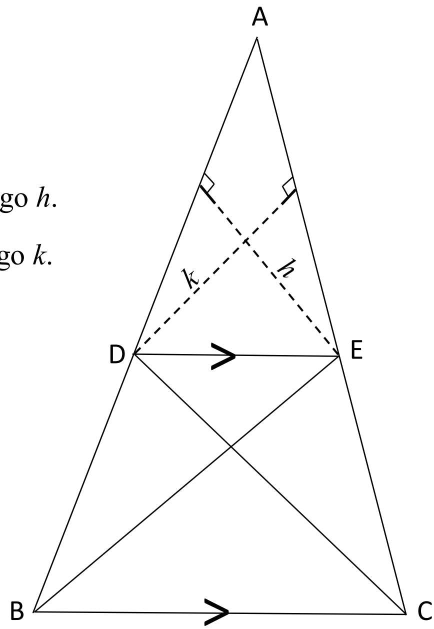
$\frac{AD}{DB} = \frac{AE}{EC}$ (Prop theorem ; DE II BC)

$\frac{AB}{AD} = \frac{AC}{AE}$ (Prop theorem ; DE II BC)

Proportionality Theorem Proof

Given $\triangle$ ABC with DE $\parallel$ BC . Prove that $\frac{AD}{DB} = \frac{AE}{EC}$

Construction :



●Dweba ulayini osuka ku B uya ku E.

●Dweba ulayini osuka ku C uya ku D.

●Dweba I height ezoya ku Base AD uyibize ngo *h*.

●Dweba I height ezoya ku Base AE uyibize ngo *k*.

Proof :

$$\frac{\text{Area } \triangle ADE}{\text{Area } \triangle BDE} = \frac{\frac{1}{2} AD \times h}{\frac{1}{2} DB \times h} = \frac{AD}{DB}$$

$$\frac{\text{Area } \triangle ADE}{\text{Area } \triangle CED} = \frac{\frac{1}{2} AE \times k}{\frac{1}{2} EC \times k} = \frac{AE}{EC}$$

Area $\triangle$BDE = $\triangle$CDE (Same Base DE and DE is //)

According to the property of triangles, the triangles drawn between the same parallel lines and on the same base have equal areas.

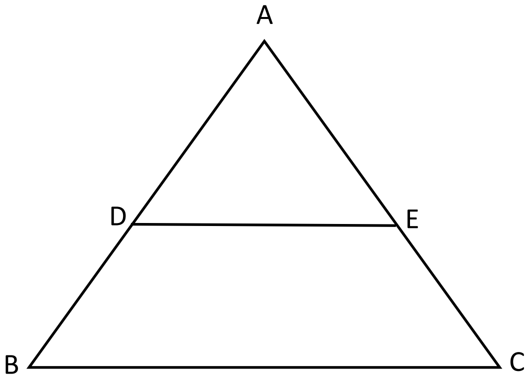
$$\frac{\text{Area}\triangle ADE}{\text{Area } \triangle BDE} = \frac{\text{Area}\triangle ADE}{\text{Area } \triangle CED}$$

$$\mathbf{\frac{AD}{DB} = \frac{AE}{EC}}$$

Converse Theorem

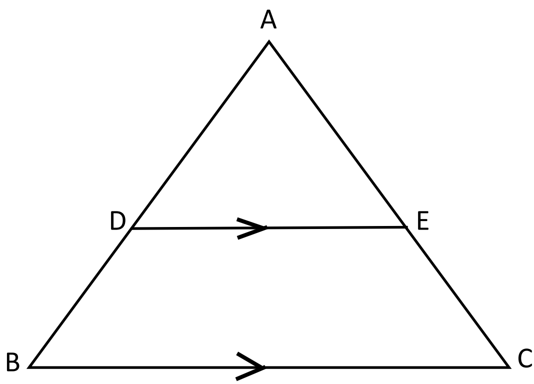
If a line cuts two sides of a triangle proportionally, then that line is parallel to the third side.

IF :



$$\frac{AD}{DB} = \frac{AE}{EC}$$

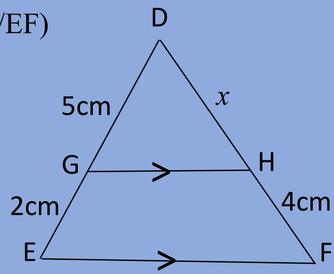
THEN :



DE || BC

IZIBONELO:

1) In Δ DEF, GH // EF. Calculate the value of x.



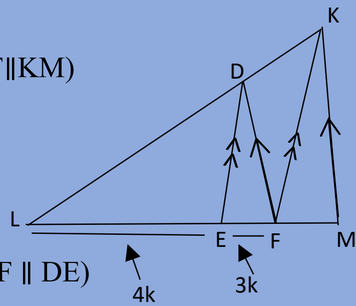
$\frac{DG}{GE} = \frac{DH}{HF}$ (Prop Theorem ; GH//EF)

$\frac{5}{2} = \frac{x}{4}$

$\frac{20}{2} = x$

$x = 10cm$

2) In Δ KLM, KM ∥ DF, KF ∥ DE and FE : EL = 3 : 4 .
Determine LE : FM.



$\frac{KD}{DL} = \frac{MF}{FL}$ (Prop Theorem;DF∥KM)

$\frac{KD}{DL} = \frac{MF}{7k}$

$\therefore \frac{FE}{EL} = \frac{KD}{DL}$ (Prop Theorem; KF ∥ DE)

$\frac{3k}{4k} = \frac{MF}{7k}$

$\frac{21k^{\cancel{2}}}{4\cancel{k}} = MF$

$\frac{21k}{4} = MF$

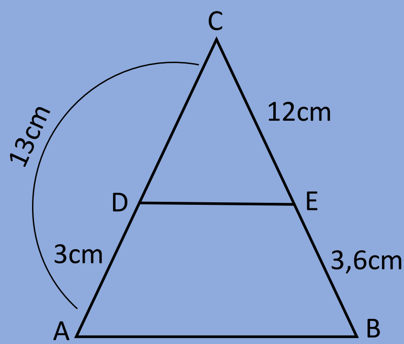
$\therefore \frac{LE}{FM} = 4k \div \frac{21k}{4}$

$= 4\cancel{k} \times \frac{4}{21\cancel{k}} = \frac{16}{21}$

∴ LE : FM = 16 : 21

NB . Isizathu esenza ukuthi siqambe u **EL** ngo **4k** no **FE** ngo **3k** ingoba sizama ukucacisa ukuthi lezyazinombolo zimele ama ratios , Azimele I Length uqobo.

In ΔABC, AC = 13 cm, AD = 3 cm; BE = 3,6 cm and EC = 12cm. Prove that DE ∥ AB.



DC = AC – AD

= 13 – 3

= 10cm

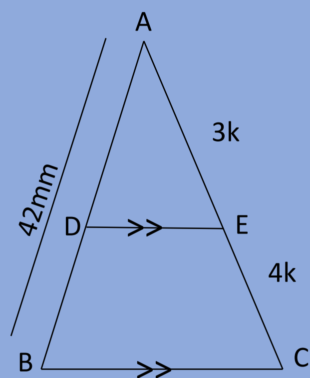
$\frac{BE}{EC} = \frac{3,6}{12} = \frac{3}{10}$

$\frac{AD}{DC} = \frac{3}{10}$

$\therefore \frac{BE}{EC} = \frac{AD}{DC}$

∴ DE ∥ AB (Line divides two sides of triangle proportionally)

3) In Δ ABC, DE ∥ BC, AB = 42 mm and AE : EC = 3 : 4. Determine the length of BD.



$\frac{BD}{AB} = \frac{EC}{AC}$ (Prop Theorem;DE∥BC)

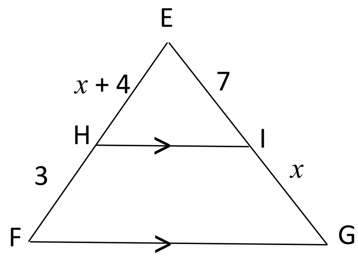
$\frac{BD}{42} = \frac{4\cancel{k}}{7\cancel{k}}$

$BD = \frac{168}{7}$

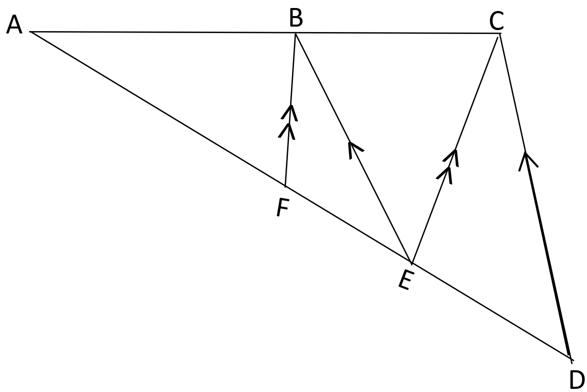
BD = 24mm

UMSEBENZI WASEKHAYA

1)Given ΔEFG with HI∥FG. Calculate the length of sides labelled x.



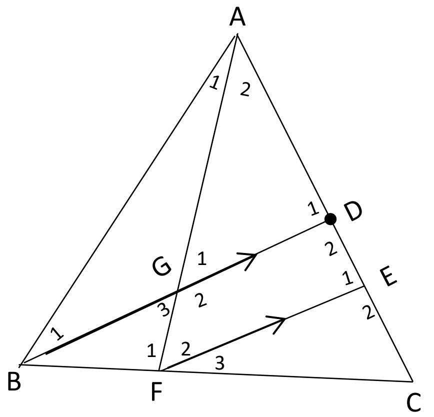
2) In ΔADC , E is a point on AD and B is a point on AC such that EB∥DC. F is a point on AC such that FB∥EC . It is given that AB = 2BC.



2.1 Determine the value of AF:FE

2.2 Calculate the length of ED if AF = 8 cm

3) In the diagram below , D is the midpoint of AC and BD||FE . BF:BC = 1: 3

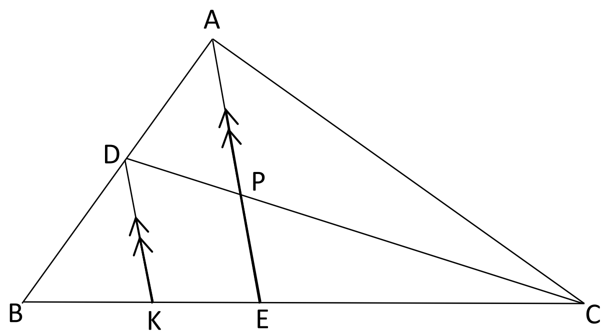


3.1 Determine , with reason , $\frac{AD}{AE}$

3.2 Prove that ΔAGD|||ΔAFE

3.3 Hence , determine $\frac{FE}{GD}$, if $\frac{AD}{AE} = \frac{3}{4}$

4) AD:BD = 2 : 3 and BC:EC = 7:3 . KD||EPA



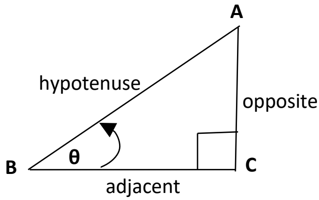
4.1 Calculate : $\frac{CP}{PD}$

Chapter 5

TRIGONOMETRY

Trigonometry : Recap

I Trigonometry isifundo esibhekise ekutholeni ubudlelwano phakathi kwama sides kanye nama angles e Triangle , Sibhekise kakhulu ekusebenziseni ama ratios ama thathu okuwu **Sin** , **Cos** kanye no **Tan** . Ebangeni leshumi nanye sike safunda lokhu nge Trigonometry :



$\text{Sin}\theta = \dfrac{\text{opposite}}{\text{hypotenuse}}$

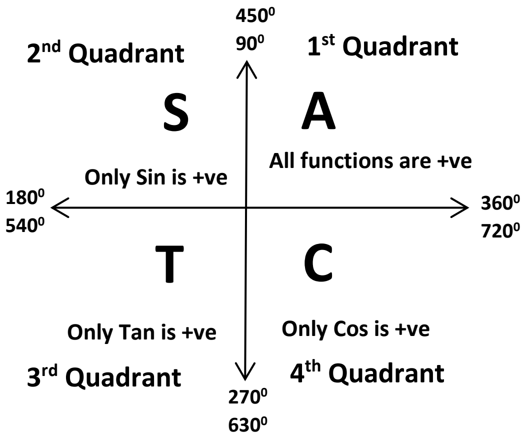
$\text{Cos}\theta = \dfrac{\text{adjacent}}{\text{hypotenuse}}$ **SOHCAHTOA**

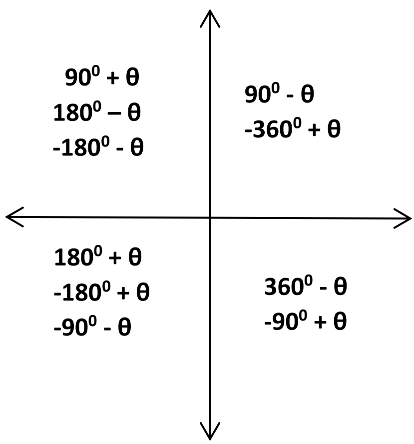
$\text{Tan}\theta = \dfrac{\text{opposite}}{\text{adjacent}}$

TRIG DIAGRAMS

***Negative Angles**

$\text{Sin}(-\theta) = -\text{Sin}\theta$
$\tan(-\theta) = -\tan\theta$
$\text{Cos}(-\theta) = \text{Cos}\theta$

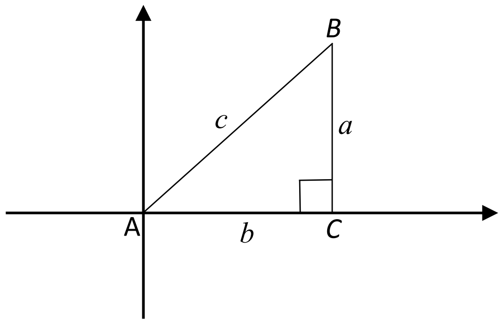




NB. Uma kusetshenziswe I degree ekuma y-axis okungaba u <u>90⁰</u> noma u <u>270 ⁰</u> kumele I trig function yakho ishintshe. Uma beku <u>Sin</u> kumele ibe u <u>Cos</u> , ngokunjalo futhi uma beku <u>Cos</u> kumele ibe u <u>Sin</u>.

i.e $\text{Sin}(90^0 + \theta) = \text{Cos}\theta$

GRADE 11 : DERIVING IDENTITIES



$$\text{SinA} = \frac{\text{opp}}{\text{hyp}} = \frac{a}{c}$$

$$\text{CosA} = \frac{\text{adj}}{\text{hyp}} = \frac{b}{c}$$

$$\begin{aligned}
\text{Sin}^2A + \text{Cos}^2A &= \left(\frac{a}{c}\right)^2 + \left(\frac{b}{c}\right)^2 \\
&= \frac{a^2}{c^2} + \frac{b^2}{c^2} \\
&= \frac{a^2 + b^2}{c^2} \qquad \mathbf{a^2 + b^2 = c^2 \ldots \text{Pythagorus.}} \\
&= \frac{c^2}{c^2} \\
&= 1
\end{aligned}$$

$$\begin{aligned}
\frac{\text{SinA}}{\text{cosA}} &= \frac{a}{c} \div \frac{b}{c} \\
&= \frac{a}{\cancel{c}} \times \frac{\cancel{c}}{b} \\
&= \frac{a}{b} \\
&= \text{tanA}
\end{aligned}$$

$\therefore \text{tanA} = \dfrac{\text{SinA}}{\text{CosA}}$

Basic Identities

- $\text{Sin}^2A + \text{Cos}^2A = 1$
- $\text{Cos}^2A = 1 - \text{Sin}^2A$
- $\text{Sin}^2A = 1 - \text{Cos}^2A$
- $(1 - \text{CosA})(1+\text{CosA}) = 1-\text{Cos}^2A$
- $(1- \text{SinA})(1+\text{SinA}) = 1-\text{Sin}^2A$
- $\tan\theta = \dfrac{\text{Sin}\theta}{\text{Cos}\theta}$
- $\dfrac{1}{\tan\theta} = \dfrac{\text{Cos}\theta}{\text{Sin}\theta}$

Examples :

Simplify : $\dfrac{1-\cos\theta}{\sin^2\theta}$ $= \dfrac{1-\cos\theta}{1-\cos^2\theta}$ $= \dfrac{1-\cos\theta}{(1-\cos\theta)(1+\cos\theta)}$ $= \dfrac{1}{1+\cos\theta}$	Prove the following identity : $\dfrac{\sin a - \sin a\cos a}{\cos a - (1-\sin^2 a)} = \tan a$ LHS $= \dfrac{\sin a - \sin a\cos a}{\cos a - (1-\sin^2 a)}$ $= \dfrac{\sin a - \sin a\cos a}{\cos a - (\cos^2 a)}$ $= \dfrac{\sin a(1-\cos a)}{\cos a(1-\cos a)}$ $= \tan a$
Prove the following : $\dfrac{1}{\text{Cos}^2\theta} - 1 = \tan^2\theta$ LHS $= \dfrac{1}{\text{Cos}^2\theta} - 1$ $= \dfrac{1-\text{Cos}^2\theta}{\text{Cos}^2\theta}$ $= \dfrac{\text{Sin}^2\theta}{\text{Cos}^2\theta}$ $= \tan^2\theta$	Prove the following : $\dfrac{\sin^2 x \,.\, \cos x \,.\, \tan x}{1-\cos^2 x} = \sin x$ LHS $= \dfrac{\sin^2 x \,.\, \cos x \,.\, \tan x}{1-\cos^2 x}$ $= \dfrac{\sin^2 x \,.\, \cancel{\cos x} \,.(\sin x/\cancel{\cos x})}{1-\cos^2 x}$ $= \dfrac{\cancel{\sin^2 x} \,.\, \sin x}{\cancel{\sin^2 x}}$ $= \sin x$
Reduction Formulae	
Simplify : $\dfrac{\cos(-\theta).\cos(90^\circ+\theta).\tan(\theta+180^\circ)}{\tan(360^\circ-\theta).\cos\theta.\sin(360^\circ+\theta)}$ $= \dfrac{(\cos\theta).(-\sin\theta).(\tan\theta)}{(-\tan\theta).(\cos\theta).(\sin\theta)}$ $= 1$	Simplify : $\dfrac{2\sin(180^\circ - x).\cos(360^\circ - x)}{\cos(90^\circ - x).\cos(180^\circ + x)}$ $= \dfrac{2\sin x\cos x}{\sin x.(-\cos x)}$ $= \dfrac{2}{-1}$ $= -2$

SPECIAL ANGLES

$\sin 30^\circ = \frac{1}{2}$	$\sin 45^\circ = \frac{1}{\sqrt{2}}$	$\sin 60^\circ = \frac{\sqrt{3}}{2}$
$\cos 30^\circ = \frac{\sqrt{3}}{2}$	$\cos 45^\circ = \frac{1}{\sqrt{2}}$	$\cos 60^\circ = \frac{1}{2}$
$\tan 30^\circ = \frac{1}{\sqrt{3}}$	$\tan 45^\circ = 1$	$\tan 60^\circ = \frac{\sqrt{3}}{1}$

Kubalulekile ukuba lama Special Angles uwazi ngoba uzowasebenzisa kakhulu

Uma udinga ukwehlisa inani lama trig function akho ***(sin , cos , tan)*** futhi uzowasebenzisa kakhulu kwama **Compound and Double** Angles okuyisfundo esiqhubeka kwi Trigonometry sika Matikuletsheni.

GRADE 12 IDENTITIES

COMPOND ANGLE IDENTITIES

- $\cos(a - \beta) = \cos a.\cos\beta + \sin a.\sin\beta$
- $\cos(a + \beta) = \cos a.\cos\beta - \sin a.\sin\beta$
- $\sin(a - \beta) = \sin a.\cos\beta - \cos a.\sin\beta$
- $\sin(a + \beta) = \sin a.\cos\beta + \cos a.\sin\beta$

DOUBLE ANGLE IDENTITIES

- $$\cos 2a = \begin{cases} \cos^2 a - \sin^2 a \\ 1 - 2\sin^2 a \\ 2\cos^2 a - 1 \end{cases}$$

$$\left\{ \begin{array}{l} \bullet \sin 2a = 2\sin a\cos a \\ \sin\tfrac{1}{2}a\cos\tfrac{1}{2}a = \tfrac{1}{2}\sin(2\times\tfrac{1}{2}a) \end{array} \right.$$

COMPOUND ANGLES EXAMPLES

a) $\cos(a + 10^\circ)$

$= \cos a.\cos 10^\circ - \sin a.\sin 10^\circ$

b) $\sin(x + y)$

$= \sin x.\cos y + \cos x.\sin y$

c) $\cos(x + 60^\circ)$

$= \cos x.\cos 60^\circ - \sin x.\sin 60^\circ$

$= \cos x.\frac{1}{2} - \sin x.\frac{\sqrt{3}}{2}$

$= \frac{\cos x - \sqrt{3}\sin x}{2}$

d) $\cos 4x.\sin 5x - \sin 4x.\cos 5x$

$= \sin 5x.\cos 4x - \sin 4x.\cos 5x$

$= \sin(5x - 4x)$

$= \sin x$

e) $\cos 75^\circ$

$\cos 75^\circ = \cos(30^\circ + 45^\circ)$

$= \cos 30^\circ.\cos 45^\circ - \sin 30^\circ.\sin 45^\circ$

$= \frac{\sqrt{3}}{2}.\frac{\sqrt{2}}{2} - \frac{1}{2}.\frac{\sqrt{2}}{2}$

$= \frac{\sqrt{6} - \sqrt{2}}{4}$

f) $\sin 70^\circ\sin 10^\circ + \cos 10^\circ\cos 70^\circ$

$= \cos 70^\circ.\cos 10^\circ + \sin 70^\circ\sin 10^\circ$

$= \cos(70^\circ - 10^\circ)$

$= \cos(60^\circ)$

$= \frac{1}{2}$

NB. 1)Uma unikezwe I nombolo eyi Special Angle kanye ne variable kwama brackets e trig function yakho kubalulekile Ukuthi uyi simplify uze ufike ekugcineni *(c).*

2)Kubalulekile ukusebenzisa ama Special Angles akho uma unikezwe inombolo ezimele futhi engagcini ngokuzimela kodwa ekwaziyo ukukhandwa ama special angles akho (i.e 75 = 45 + 30) *(e).*

DOUBLE ANGLES EXAMPLES

Lesisifundo siqhubekela olwazini lwaka Grade11 loku solve ama trigonometric problems kepha kulelibanga sisuke sesandisa ulwazi lwakho ngokufaka ama formula amabili abhaliwe ngenhla.

a) $2\sin15^{o}\cos15^{o}$

$= \sin(2\times15^{o})$ . . .**sin2*a* = 2sin*a*cos*a***

$= \sin30^{o}$

$= \dfrac{1}{2}$

b) $\cos6x = \cos(2 \times 3x)$ . . . **cos2a**

$= \cos^2 3x - \sin^2 3x$ **or**

$= 1 - 2\sin^2 3x$ **or**

$= 2\cos^2 3x - 1$

c) $\sin8x$

$= \sin(2\times4x)$

$= 2\sin4x\cos4x$

d) $\cos20x = \cos(2\times10x)$

$= \cos^2 10x - \sin^2 10x$ **or**

$= 1 - 2\sin^2 10x$ **or**

$= 2\cos^2 10x - 1$

e) $\sin12x$

$= \sin(2\times6x)$

$= 2\sin6x\cos6x$

f) $\cos120^{o} = \cos(2\times60^{o})$

$= \cos^2 60^{o} - \sin^2 60^{o}$

$(1/2)^2 - (\sqrt{3}/2)^2$

$(1/4) - (3/4)$

$(-2/4)$

$= (-1/2)$ **or**

$= 1 - 2\sin^2 60^{o}$

$1 - 2(\sqrt{3}/2)^2$

$1 - 2\ (3/4)$

$1 - (3/2)$

$= (-1/2)$ **or**

$= 2\cos^2 60^{o} - 1$

$2\ (1/2)^2 - 1$

$2(1/4) - 1$

$(1/2) - 1$

$= (-1/2)$

g) $2\sin3x\cos3x$

$= \sin(2\times3x)$

$= \sin6x$

h) $4\sin22{,}5\cos22{,}5$

$= 2.2\sin22{,}5\cos22{,}5$

$= 2.\sin(2\times22{,}5)$

$= 2\sin(45)$

$= \cancel{2}.\dfrac{\sqrt{2}}{\cancel{2}}$

$= \sqrt{2}$

i) **$\sin15^{o}\cos15^{o}$**

$= ½ \sin(2 \times 15^{o})$

$= ½\times½$

$= ¼$

Uma Isbalo onikezwe sona ubona okuthi siyi formula ka sin2a kodwa asiqali ngo 2 ekuqaleni . sike sisebenzise le formula sin½acos½a = ½sin(2×½a)

UMSEBENZI WASEKHAYA

(Compund and Double Angles , Reduction formulae , Proving Identities)

1)Simplify the following : a) $\dfrac{\sin 200^\circ.\cos 160^\circ}{\sin 10^\circ.\cos 30^\circ + \sin 30^\circ \cos 10^\circ}$

b) $\dfrac{\sin 2x}{\sin x} - \dfrac{\cos 2x}{\cos x}$

c) $\sin 140^\circ.\cos 330^\circ - \cos 320^\circ \sin 150^\circ$

d) $\cos 290^\circ.\cos 330^\circ - \sin 250^\circ.\sin 210^\circ$

e) $\cos 285^\circ.\sin 75^\circ + \sin 105^\circ.\sin 15^\circ$

2) Calculate the following without using a calculator :

$\text{Sin}236^\circ.\cos 169^\circ + \sin 371^\circ.\cos(-124^\circ)$

3) simplify the following :

$\dfrac{\tan(-x).\cos^2(180^\circ - x)}{\sin(360^\circ - 2x)}$

4) $\dfrac{\cos^2(-x) + \cos^2(90 + x) - \cos(180 + 2x)}{\text{Sin}^2(90 + x)}$

5) $\dfrac{\cos^2 173 - \sin^2 7}{\sin(-66).\cos 190 - \cos 66.\sin 350}$

6) Prove the following Identities : a) $\dfrac{\sin x}{1-\sin x} + \dfrac{\sin x}{1+\sin x} = \dfrac{2\sin x}{\cos^2 x}$

b) $\tan x = \dfrac{\sin 2x}{1 + \cos 2x}$

c) $\dfrac{\sin x - \cos 2x}{\sin 2x - \cos x} = \dfrac{\sin x + 1}{\cos x}$

GR11: CAST DIAGRAM

Lesi isifundo esibheke ekudwebeni I Right Angle Triangle kweyodwa yama Quadrants kwi Cartesian Plane yakho ebese usebenzisa umdwebo wakho ukwenza izibalo/calculations. Ukukhetha ukuba uzodweba kuyiphi I Quadrant kulele kwi problem ozobe unikezwe yona kanti kusemqoka ukuba uyihlolisise ukuze uzophuma nezimpendulo ezifanele. Nayi imibuzo okumele uzibuze yon muthola I problem :

Given $5\cos\theta = 4$ and $180 < \theta < 360$

$\therefore$ Determine $\cos 2\theta$

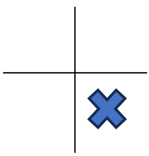
Step 1 : Lungisa I equation onikezwe yona , esandleni sokunxele kusale I trig function kuphela mase esandleni sokudla kuba izinombolo zodwa.

$$*\cos\theta = \frac{4}{5}$$

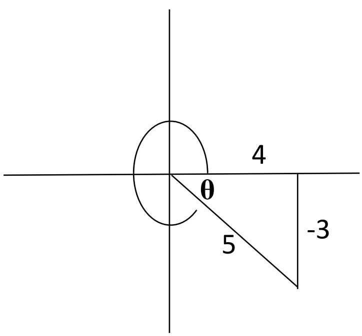
$$\cos\theta = \frac{Adj}{Hyp} = \frac{4}{5}$$

Umbuzo. *$180 < \theta < 360$ and $+4$* : iyiphi i Quadrant engaphezulu kuka 180 kodwa futhi engaphansi kuka 360 lapho khona u cos e +ve(+*4*) ?

Answer :



Drawing :



$x^2 + y^2 = r^2$

$4^2 + y^2 = 5^2$

$y = \pm\sqrt{9}$

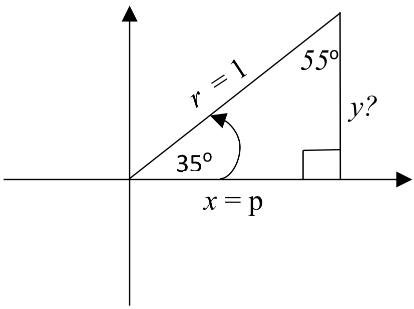
$y = -3$

$$\therefore \cos 2\theta = 2\cos^2\theta - 1$$
$$= 2(4/5)^2 - 1$$
$$= 2(16/25) - 1$$
$$= (7/25)$$

GRADE 12 CAST DIAGRAM

Given : $\cos 35^{o} = p$

1st Step. Sebenzisa I equation onikezwe yona ukudweba I Right Angled Triangle yakho , ebese wenza izibalo zokuthola I third angle.



$$Cos35^{o} = \frac{Adj}{Hyp} = \frac{p}{1}$$

$35^{o} + 90^{o} + x = 180$ … Sum of Angles of Triangle = 180

$x = 55^{o}$

2nd Step. Sebenzisa I Pythagoras Theorem ukuthola inani lesayidi lesithathu leli elingakabi ne value *(kulesisimo isayidi la **y** esingenayo i value yalo).*

$$\boldsymbol{x^2 + y^2 = r^2}$$

$$\boldsymbol{(p)^2 + y^2 = (1)^2}$$

$$\boldsymbol{y = \sqrt{1 - p^2}}$$

3rd Step. Emvakokwenza konke lokhu okungenhla uzobe suthola izibalo okuzomele uzibalele . I lapho sekuzomele usebenzise ulwazi lwama Special , Double and Compound angles uphinde ukwazi ukuhlolisisa ukuba le angle onikezwe yona okuba yibalele ingakhandeka kanjani ngokusebenzisa ama angles akwi Right Angled Triangle yakho.

Determine the following in terms of P.

$$\sin 35^\circ = \frac{Opp}{Hyp} = \frac{y}{r}$$

$$= \frac{\sqrt{\mathbf{1}-\boldsymbol{p}^2}}{1}$$

$$= \sqrt{\mathbf{1}-\boldsymbol{p}^2}$$

$\tan 215^\circ + \sin(-55^\circ)$

$= \tan(180 + 35^\circ) + (-\sin 55^\circ)$

$= (\tan 35^\circ) + (-\sin 55^\circ)$

$$= \left(\frac{\sqrt{\mathbf{1}-\boldsymbol{p}^2}}{p}\right) + \left(\frac{-p}{1}\right)$$

$$= \frac{\sqrt{1-p^2} - p^2}{p}$$

$\sin 70^\circ = 2\sin 35^\circ \cos 35^\circ$

$$= 2.\left(\frac{\sqrt{1-p^2}}{1}\right).\, p$$

$$= 2p\sqrt{\mathbf{1}-\boldsymbol{p}^2}$$

$\cos 20^\circ = \cos(55^\circ - 35^\circ)$

$= \cos 55.\cos 35 + \sin 55.\sin 35$

$$= \frac{\sqrt{1-p^2}}{1}.\frac{p}{1} + \frac{p}{1}.\frac{\sqrt{1-p^2}}{1}$$

$= p\sqrt{1-p^2} + p\sqrt{1-p^2}$

$= 2\,(p\sqrt{1-p^2}\,)$

$= 2p\sqrt{1-p^2}$

General Solution

$\text{Sin}\theta = (\tfrac{1}{2})$ $\theta = 30^o$ $\theta = 30^o + 360^o.k$ *OR* $\theta = 180 - 30^o + 360.k$ $= 150 + 360.k\ ,\ k \in \mathbb{Z}$	**Uma unikezwe I trig function esandleni sokunxele wase unikezwa i nombolo esandleni sokudla , kubalulekile ukuthi usebenzise I calculator yakho ukuthola I reference angle.**
$2\sin\theta = 3\cos\theta$ $\dfrac{\sin\theta}{\cos\theta} = \dfrac{3}{2}$ **. . . (sin/cos) = tan** $\tan\theta = \dfrac{3}{2}$ $\theta = 56{,}31^o + 180^o.k\ ,\ \ k \in \mathbb{Z}$ *OR* $\theta = 180^o + 56{,}31 + 180^o.k$ $= 236{,}31 + 180^o.k\ ,\ \ k \in \mathbb{Z}$	**Kubalulekile ukuthi ungawakhohlwa ama Basic Identities abefundiwe ngaphambilini ngoba usazolidinga ulwazi lawo ukuqhamuka nezimpendulo ezifanele. i.e($^{\sin}/_{\cos}$) = tan**
$\cos\theta = \cos(45 + b\ )$ $\theta = \pm\,(45 + b\ ) + 360^o.k\ ,\ \ k \in \mathbb{Z}$	**Uma unikezwe ama trig function afanayo ndawo zombili (i.e cos = cos)kubalulekile ukuthi wehlise ama angle njengoba enjalo .**
$\sin\theta = \cos a$ $\sin\theta = \sin(90 - a\ )$ $\theta = 90 - a\ .\ .\ .\ ref <$ $\theta = 90 - a + 360^o.k\ ,\ \ k \in \mathbb{Z}$ *OR* $\theta = 180 - (90 - \text{a}) + 360.k$ $\theta = 90 + a + 360^o.k\ ,\ \ k \in \mathbb{Z}$	**Uma unikezwe ama trig function angafani ndawo zombili , aphinde futhi angabi nama angle afanayo . kubalulekile ukuthi ujike eyodwa I trig function ukuze ifane nenye.**

Kuyenzeka okuba I equation onikezwe yona ingafani nalezindlela ezine ezingenhla .Uma uficana nesimo esinjalo kubalulekile ukuthi usebenzise ulwazi obeselufundiwe lwama identities , compound and double angles uku Factorise.

UMSEBENZI WASEKHAYA

Determine The General Solutions For :

1)Cos2b = - 0,174° for -180° $\leq b \leq$ 180°

2) $\sin(3\theta + 50^o) + \cos(2\theta - 10^o) = 0$. Where -180° $\leq \theta \leq$ 180°

3) $2\tan x - 3 = \cos 32^o$ for [0°; 180°]

4) $\sin 2x = (0,5)$ for $x \in [0^o; 90^o]$

5) $3\cos 2x = - 2,34$ for [0°; 180°] . correct to two decimal places .

6) $\cos^2 x + \sin 2x - 1 = 0$.

7) $\tan x \cdot \sin x + \cos x = \dfrac{3}{\sin x}$

8) $\sin 3x \cos x - \cos 3x \sin x = \sin x$

9) $3\sin x \cdot \sin 22^o = 3\cos x.\cos 22^o + 1$

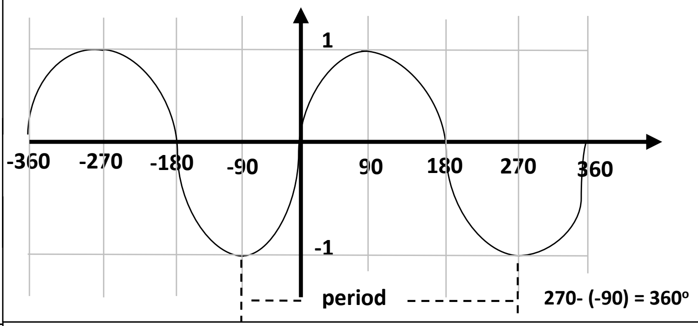
10) $4\sin^2 x + 6\sin x.\cos x - 2\sin x - 3\cos x = 0$ for -360° $\leq x \leq$ 360° . Round off the answer to 2 decimal places if necessary.

11) $2\sin 2x + \cos 2x + 2 = 0$.

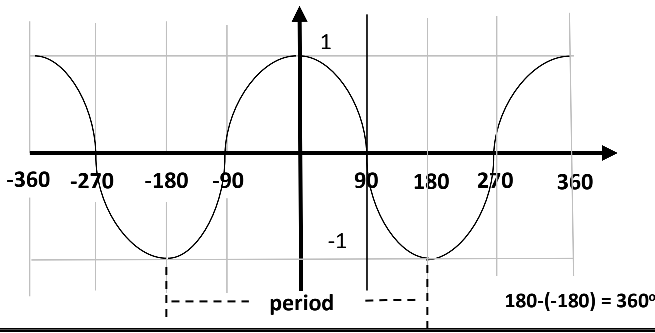
TRIGONOMETRIC GRAPHS

Basic Trigonometric Graphs

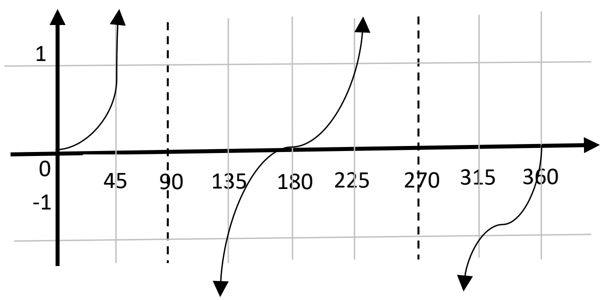
y = sin$x$
-360º ≤ x ≤ 360º



y = cos$x$
-360º ≤ x ≤ 360º



y = tan$x$
0 ≤ x ≤ 360º



PERIOD	AMPLITUDE
sin$x$ = 360º	1
cos$x$ = 360º	1
tan$x$ = 180º	1

PERIOD – Isho ibanga elithathekile ukuthi I Cycle yokuqala ye Trigonometric Graph yenzeke / kusitshela Ibanga amagquma ethu womabili aqhelelene ngalo.

AMPLITUDE – Kuchaza ukunyuka nokwehla kwe graph yethu ngama y – axis ***(AMPLITUDE ihlezi I Positive).***

Kumele ukwazi ukudweba ama Trigonometric Graphs akho futhi uphinde ukwazi ukwahlolisisa uqhamuke nezimpendulo ezifanele . Kuke kwenzeke lama Graphs angenhla bawajije/manipulate , Lapho besuke bejije I Period, Amplitude noma bayi Shift I Graph yakho. Nayi indlela okuke bakwenze ngakho lokhu :

- $y = a\sin k(x + p) + q$
- $y = a\cos k(x + p) + q$
- $y = a\tan k(x + p) + q$

a	Affects the Amplitude
k	Affects the Period For cos and sin : $\frac{360}{k}$ For tan : $\frac{180}{k}$
$(x + p)$	Shifts the graphs horizontally
q	Shifts the graphs vertically

IZIBONELO :

GRAPH	**PERIOD**	**AMPLITUDE**	**RANGE**
$y = 2\sin\theta$	$\frac{360^o}{1} = 360^o$	2	$-2 \leq y \leq 2$
$y = -4\sin2\theta$	$\frac{360}{2} = 180^o$	4	$-4 \leq y \leq 4$
$y = 4\cos\frac{3}{4}\theta$	$\frac{360}{\frac{3}{4}} = 480^o$	4	$-4 \leq y \leq 4$

2D AND 3D PROBLEMS

Lesi isifundo esibheke ekuxazululeni ama sides kanye nama angles asuke ekwi triangle , okusemqoka ukuba le triangle ayisiyo I Right Angled Triangle. Lapha sisebenzisa ama rules amathathu okuwu Sine , Cosine and Area Rule. Nakhu okumele sikwazi ngalama Rules :

Sine RULE : $\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$

- **Uma unikezwe ama sides amabili kanye ne angle ebhekene ne side elilodwa kulawa amabili , ungayisebenzisa le rule ukubalela i angle yaleli elinye i side.**
- **Uma unikezwe ama angles amabili kanye ne side elibhekene neyodwa ye angle kulawa amabili , ungayisebenzisa le rule ukubalela i side yale enye i angle.**

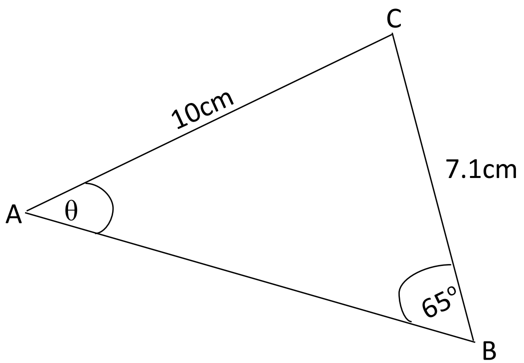
Cosine Rule : $c^2 = a^2 + b^2 - 2ab\cos C$

- **Uma unikezwe ama sides amabili kanye ne angle ephakathi kwawo womabili lama sides , ungayisebenzisa le rule ukubalela i side elibhekene ne angle.**
- **Uma unikezwe ama sides amathathu , ungayisebenzisa le Rule ukubalela i angle engaziwa.**

Area Rule : $\frac{1}{2}ab\sin C$

- **Uma unikezwe ama sides amabili kanye ne angle ephakathi kwawo womabili lama sides , ungayisebenzisa le rule ukubalela i Area ye Triangle .**

Sine Rule



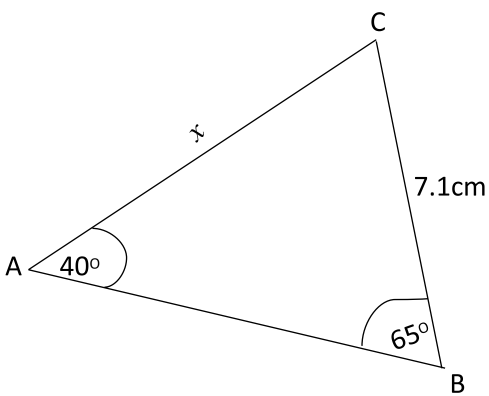
$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

$$\frac{\sin\theta}{7.1} = \frac{\sin 65^\circ}{10}$$

$$\sin\theta = \frac{7.1\sin 65^\circ}{10}$$

$$\theta = \sin^{-1}(0,6434785288)$$

$$\theta = 40,05^\circ$$



$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

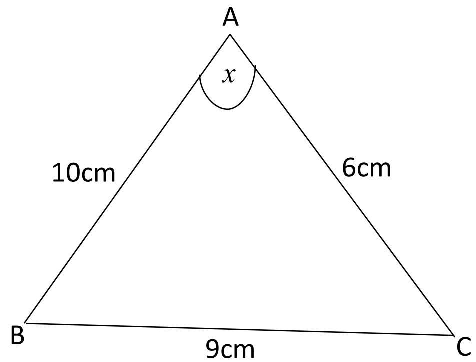
$$\frac{\sin 40^\circ}{7.1} = \frac{\sin 65^\circ}{x}$$

$$x = \frac{7.1\sin 65^\circ}{\sin 40^\circ}$$

$$= 10,01\text{cm}$$

$$x = 10\text{cm}$$

Cosine Rule :



$c^2 = a^2 + b^2 - 2ab\cos C$

$(9)^2 = (6)^2 + (10)^2 - 2(6)(10)\cos x$

$81 = 36 + 100 - 120\cos x$

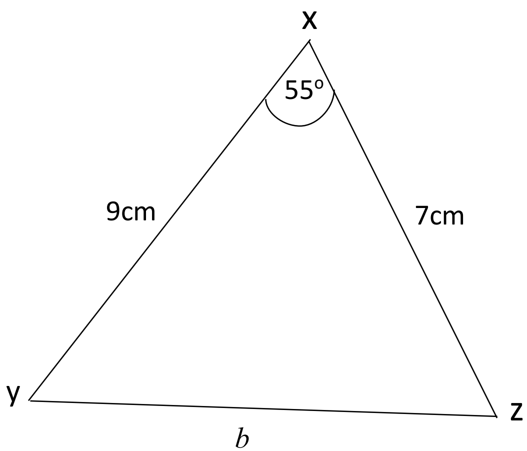
$81-36-100 = -120\cos x$

$-55 = -120\cos x$

$0{,}458 = \cos x$

$\cos^{-1}(0{,}458) = x$

$x = 62{,}72^{\circ}$



$c^2 = a^2 + b^2 - 2ab\cos C$

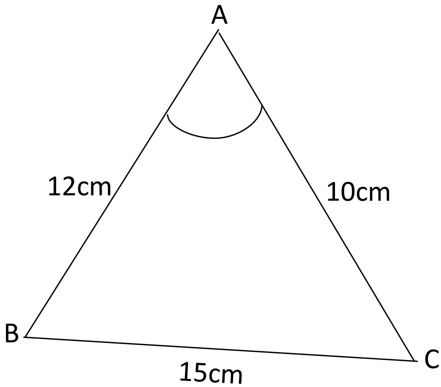
$b^2 = (7)^2 + (9)^2 - 2(7)(9)\cos 55^{\circ}$

$b^2 = 49 + 81 - 72{,}27$

$b = \sqrt{57{,}73}$

$b = 7{,}6\text{cm}$

Area Rule :



$c^2 = a^2 + b^2 - 2ab\cos C$

$15^2 = 10^2 + 12^2 - 2(10)(12)\cos C$

$225 = 100 + 144 - 240\cos C$

$-19 = -240\cos C$

$0,0792 = \cos C$

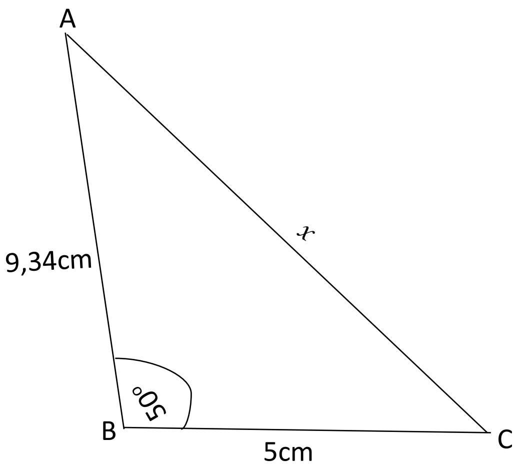
$\cos^{-1}(0,0792) = C$

$C = 85,46^{o}$

Area $\triangle ABC = \frac{1}{2}ab\sin C$

$= \frac{1}{2}.10.12\sin 85,46$

$= 59,81cm^2$



Area $\triangle ABC = \frac{1}{2}ab\sin C$

$= \frac{1}{2}.9,34.5\sin 50^{o}$

$=17,89cm^2$

Chapter 6

CALCULUS

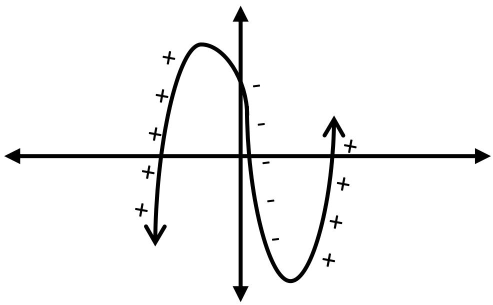
Cubic Function

I Calculus isifundo esibheke kakhulu kuma gradients kanti sinohlobo lwe graph esike silubize nge **Cubic Function** , Lolu uhlobo lwe graph oluba nama *turning points* **amabili** liphinde linqamule ama *x-axis* kwizindawo ezingagcina **zintathu**. I Cubic Function iba naloluhlobo lwe Formula :

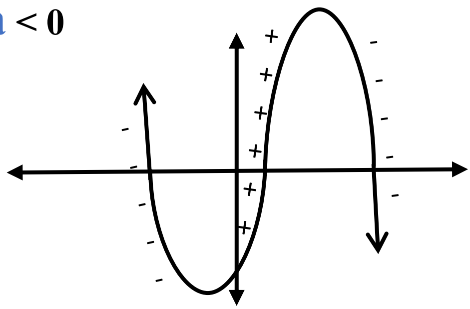
$$f(x) = \mathbf{a}x^3 + \mathbf{b}x^2 + \mathbf{c}x + \mathbf{d}$$

Zimbili izindlela zokudweba I Cubic Function kanti ukukhetha indlela okumele I Function yakho idwebeke ngayo lokho kwenziwa u **a**. Nakhu okumele sikwazi ngo **a** :

Uma u a > 0



Uma u a < 0



First Derivative

Mhlawumbe uyazibuza uyaziphendula ukuba siwathola kanjani ama ***turning points*** kanye nama ***x-intercepts*** , Lokho kulele ekutheni siqale sifunde nge First Derivate okuyisifundo esisemqoka kwi Calculus. Njengoba besishilo ngenhla ukuthi isifundo se Calculus sibheke kakhulu kuma Gradients , I First Derivative iyona ndlela yokuthi sithole I Gradient yethu lapho sisebenzisa khona I equation yethu ebesiyifunde engenhla . I Gradient yethu siyithola kanje :

Uma $f(x) = x^n$, I First Derivative izoba $f'(x) = nx^{n-1}$

IZIBONELO :

Equation $f(x)$	**First Derivative/Gradient $f'(x)$**
$f(x) = x^3 - 2x^2 + 3x - 4$	$f'(x) = 3x^{3-1} - (2)2x^{2-1} + (1)3x^{1-1}$ $= 3x^2 - 4x + 3$
$f(x) = 2x^3 - 4x^2 + 4x - 16$	$f'(x) = 6x^2 - 8x + 4$
$f(x) = 5x^2 - 5x$	$f'(x) = 10x - 5$
$f(x) = 7x^3 - 10x^2 + 8x - 16$	$f'(x) = 21x^2 - 20x + 8$
$f(x) = -2x^3 - 3x^2 + 9x - 4$	$f'(x) = -6x^2 - 6x + 9$

Nge **First Derivative** iyona ndlela esithola ngayo I Gradient yethu kanti kusemqoka ukuthi wenze isiqiniseko sokuthi uyenza ngendlela ngoba iyona evamise ukuphatha ukhiye ekuphenduleni imibuzo eminingi.

Ukuthola ama x – intercepts , y – intercepts nama Turning Points.

x – intercepts : i.e $f(x) = x^3 - 3x^2 - 6x + 8$

$f(x) = x^3 - 3x^2 - 6x + 8$

$0 = x^3 - 3x^2 - 6x + 8$ . . . *1.* **Enza u y = 0**

$0 = (-1)^3 - 3(-1)^2 - 6(-1) + 8 = 10$ ✖

$0 = (0)^3 - 3(0)^2 - 6(0) + 8 = 8$ ✖

$0 = (1)^3 - 3(1)^2 - 6(1) + 8 = 0$ ✔

. *2* .

Ukuthola I ***x*-intercept** yokuqala kumele uqagele I nombolo ebese uyifaka kwi equation onikezwe yona uthole inani/isamba sayo , I nombolo oyifakile yase ikunikeza inani/isamba esingu **0** iyona nombolo eyi ***x*-intercept** yakho yokuqala. Njengoba lapha kube inombolo u **1** esinikeze inani u **0** . Kumele manje inombolo yethu siyibuyisele emumva ibe kwama ***Bracket*** :

$x = 1$

$(x - 1) = 0$ ✔

. *3* .

$0 = x^3 - 3x^2 - 6x + 8$

+8

ST.2

$0 = (x - 1)(x^2 - 2x - 8)$

ST.1

x^3

ST.3

$0 = (x-1)(x-4)(x+2)$

$x = 1$

$x = 4$

$x = -2$

co-eff x : -6

$-6 = -8 + (-1)(B)$

$-6 = -8 - B$

$2 = -B$

$-2 = B$

y – Intercept : i.e $f(x) = x^3 - 3x^2 - 6x + 8$

$f(x) = (0)^3 - 3(0)^2 - 6(0) + 8$ . . . Enza u x ube 0

$f(x) = 8$

Turning Point : i.e $f(x) = x^3 - 3x^2 - 6x + 8$

$f'(x) = 3x^2 - 6x - 6$

$$0 = 3x^2 - 6x - 6$$

$$0 = x^2 - 2x - 2$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(-2)}}{2(1)}$$

$$x = \frac{2 \pm \sqrt{4+8}}{2}$$

$$x = \frac{2 \pm \sqrt{12}}{2}$$

$x = 2{,}73$ or $x = -0{,}73$

y = -10,39 y = 10,39

A(2,73 ; -10,39) B(-0,73 ; 10,39)

Calculus : Equation Of A Tangent

Ukuthola I ***Tangent*** *ithintene ne* ***Cubic Graph*** ezinye zezimo ezike zibe khona kwi calculus . Kanti okusemqoka okumele ukwazi ukuba kwi phoyinti lapho khona I tangent ithintana ne cubic graph , I **Gradient iyalingana** kulelo phoyinti ***(Gradient Tangent = Gradient Graph).*** Ukwazi lolulwazi kuzosiza kakhulu kwizimo lapho okuzomele uthole khona I equation ye Tangent ube unikezwe I equation ye Cubic Graph.

Point Of Contact : ***Gradient Tangent = Gradient Graph***

IZIBONELO :

Determine the equation of a tangent to the graph : $f(x) = x^3 - 3x^2 - 6x - 8$ where $x = 2$.

Step1 . Kumele uqale uthole i y – coordinate ye phoyinti lakho uma bunganikeziwe , lokho uzokwenza ngokuthatha I x - coordinate uyi substitute kwi graph yakho.

$f(2) = (2)^3 - 3(2)^2 - 6(2) - 8$
$\quad = -24$
POC(2;-24)

Step2. Kumele wenze I ***first derivative*** ukuze uthole I gradient ye graph. Khumbula ukuthi kwi phoyinti lapho I graph ithintana khona ne tangent I ***gradient iyalingana*** . Ngakho ke singayisebenzisa I gradient ye graph ukuthola I inani eliyiqiniso le gradient kulelo phoyinti I tangent ne graph kuthintana khona.

$f^{/}(x) = 3x^2 - 6x - 6$
$f^{/}(2) = 3(2)^2 - 6(2) - 6$
$\quad = -6$ . . . *inani le gradient kwi point of contact*

Step3. Njengoba I formula ye tangent ithi $y = mx + c$, kumele uthole u **c** ngokuthatha ama coordinates akho kanye ne gradient ukufake kwi formula ye tangent.

$y = mx + c$

$-24 = -6(2) + c$

$c = -12$

EQUATION : $y = -6x - 12$

2.Determine the equation of a tangent to the graph : $f(x) = x^3 - 2x^2 - 3x + 4$ where $x = 3$.

$f(3) = (3)^3 - 2(3)^2 - 3(3) + 4$

$= 4$

POC(3,4)

$f'(x) = 3x^2 - 4x - 3$

$f'(3) = 3(3)^2 - 4(3) - 3$

$= 12$

$y = mx + c$

$4 = 12(3) + c$

$c = -32$

$\therefore y = 12x - 32$

Eminye imibuzo ebheke kuma Gradients ike ibuzwe ngalendlela :

INFLECTION POINT : ilapho khona I Graph ishintsha isuka ekubeni positive ibe negative noma isuka ekubeni negative ibe positive . I inflection point siyithola ngokwenza I second derivative yethu ibe u 0.

$$\boxed{f^{//} = 0}$$

IZIBONELO :

Given $f(x) = x^3 + 2x^2 + 4x - 8$, Determine the inflection point.

$f(x) = x^3 + 2x^2 + 4x - 8$

$f^{/}(x) = 3x^2 + 4x + 4$

$f^{//}(x) = 6x + 4$

$0 = 6x + 4$

$x = \frac{-2}{3}$

For which values of x will $f(x)$ **concave up** ?

$f(x) = x^3 + 2x^2 + 4x - 8$

$f^{/}(x) = 3x^2 + 4x + 4$

$f^{//}(x) = 6x + 4$

$6x + 4 > 0$

$x > \frac{-4}{6}$

$x > \frac{-2}{3}$

NB

Concave up

$F^{//}(x) > 0$

Concave Down

$F^{//}(x) < 0$

When The Gradient Is Given

The equation of a Graph is given as : $f(x) = x^3 - 3x^2 - 23x + 7$. Determine the values of x on the graph where the gradient equals to **1** ?

$f(x) = x^3 - 3x^2 - 23x + 7$

$f'(x) = 3x^2 - 6x - 23$

$\boldsymbol{1} = 3x^2 - 6x - 23$

$0 = 3x^2 - 6x - 24$

$0 = x^2 - 2x - 8$

$0 = (x - 4)(x + 2)$

x = 4 or x = -2

2. The equation of a Graph is given as : $f(x) = x^3 - 6x^2 + 8x + 9$. For which values of x is the gradient = ***-1*** ?

$f(x) = x^3 - 6x^2 + 8x + 9$

$f'(x) = 3x^2 - 12x + 8$

$-1 = 3x^2 - 12x + 8$

$0 = 3x^2 - 12x + 9$

$0 = x^2 - 4x + 3$

$0 = (x - 3)(x - 1)$

x = 3 or x = 1

Determining The First Derivative Of *f(x)* Using *First Principles*

Ziningi izindlela zokuthola I First Derivative/Gradient kanti ukusebenzisa I ***First Principle Formula*** kungenye yezindlela ezisetshenziswa kakhulu kulelibanga . Le Formula isebenzisana ne equation esisuke sinikezwe yona ukuze sikwazi ukwenza I First Derivative . Nakhu okumele sikwazi nge First Principle Formula :

$$\boxed{f^{/}(x) = \lim_{h \to 0} \frac{f(x + h) - f(x)}{h}}$$

- ***f(x)*** : I equation ka *f(x)* injengoba injalo .
- ***f(x + h)*** : I equation ka *f(x)* kodwa okuhlukile ukuthi yonke indawo ebino *x* ngaphambilini sizoyishintsha sifake u *(x + h)* kuyo.

i.*e* $\boldsymbol{f(x)} = x^2 - 3x - 4$

$\boldsymbol{f(x+h)} = (x+h)^2 - 3(x+h) - 4$

IZIBONELO :

1.Given $f(x) = x^2 - 3x - 4$. Determine the First Derivative of $f(x)$ using First Principle.

$$f^{/}(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{\overbrace{\left[(x+h)^2 - 3(x+h) - 4\right]}^{f(x+h)} - \overbrace{\left[(x^2 - 3x - 4)\right]}^{f(x)}}{h}$$

$$= \lim_{h \to 0} \frac{\cancel{x^2} + 2xh + h^2 - \cancel{3x} - 3h \cancel{-4} - \cancel{x^2} + \cancel{3x} + \cancel{4}}{h}$$

$$= \lim_{h \to 0} \frac{2xh + h^2 - 3h}{h}$$

$$= \lim_{h \to 0} \frac{\cancel{h}(2x + h - 3)}{\cancel{h}}$$

$$= \lim_{h \to 0} 2x + h - 3$$

$$= 2x + 0 - 3$$

$$= 2x - 3$$

2. Determine $f^{/}(x)$ from First Principle if it is given that $f(x) = 2x^2 - 3x$.

$$f^{/}(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{2(x+h)^2 - 3(x+h) - (2x^2 - 3x)}{h}$$

$$= \lim_{h \to 0} \frac{2(x^2 + 2xh + h^2) - 3x - 3h - 2x^2 + 3x}{h}$$

$$= \lim_{h \to 0} \frac{\cancel{2x^2} + 4xh + 2h^2 - \cancel{3x} - 3h - \cancel{2x^2} + \cancel{3x}}{h}$$

$$= \lim_{h \to 0} \frac{4xh + 2h^2 - 3h}{h}$$

$$= \lim_{h \to 0} \frac{\cancel{h}(4x + 2h - 3)}{\cancel{h}}$$

$$= \lim_{h \to 0} 4x + 2h - 3$$

$$= 4x + 2(0) - 3$$

$$= 4x - 3$$

3. Determine $f^{/}(x)$ from First Principle if it is given that $f(x) = \frac{2}{x}$

$$f^{/}(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h\to 0} \left(\frac{\frac{2}{x+h} - \frac{2}{x}}{h} \right)$$

$$= \lim_{h\to 0} \left(\frac{2x - 2(x+h)}{x(x+h)} \right) \div h$$

$$= \lim_{h\to 0} \left(\frac{\cancel{2x} - \cancel{2x} - 2h}{x(x+h)} \right) \div h$$

$$= \lim_{h\to 0} \left(\frac{-2\cancel{h}}{x^2 + xh} \right) \times \left(\frac{1}{\cancel{h}} \right)$$

$$= \lim_{h\to 0} \frac{-2}{x^2 + xh}$$

$$= \frac{-2}{x^2 + x(0)}$$

$$= \frac{-2}{x^2}$$

4. Determine $f'(x)$ from First Principle if it is given that $f(x) = \dfrac{x}{2x+1}$

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h\to 0} \left(\frac{x+h}{2(x+h)+1}\right) - \left(\frac{x}{2x+1}\right) \div h$$

$$= \lim_{h\to 0} \left(\frac{(x+h)(2x+1) - x(2(x+h)+1))}{(2(x+h)+1)(2x+1)}\right) \times \left(\frac{1}{h}\right)$$

$$= \lim_{h\to 0} \left(\frac{2x^2 + x + 2xh + h - (2x^2 + 2xh + x)}{(2(x+h)+1)(2x+1)}\right) \times \left(\frac{1}{h}\right)$$

$$= \lim_{h\to 0} \frac{\cancel{2x^2} + \cancel{x} + \cancel{2xh} + \cancel{h} - \cancel{2x^2} - \cancel{2xh} - \cancel{x}}{(2(x+h)+1)(2x+1)} \times \left(\frac{1}{\cancel{h}}\right)$$

$$= \lim_{h\to 0} \left(\frac{1}{(2(x+h)+1)(2x+1)}\right)$$

$$= \left(\frac{1}{(2(x+0)+1)(2x+1)}\right)$$

$$= \left(\frac{1}{(2x+1)(2x+1)}\right)$$

$$= \left(\frac{1}{(2x+1)^2}\right)$$

UMSEBENZI WASEKHAYA

Determine $f'(x)$ from First Principle for the following equations :

1 . $f(x) = -x^2$

2 . $f(x) = \dfrac{1}{x^3}$

3 . $f(x) = 3x - x^2$

4 . $f(x) = 2x^3$

5 . $f(x) = - x^2 -5x + 4$

6 . $f(x) = \dfrac{-3}{x}$

7 . $f(x) = x^3 - 6x$

8 . $f(x) = x^2 + 2x$

9 . $f(x) = \dfrac{x - 1}{x + 1}$

10. $f(x) = 2x^2 -1$

Sizwe sika Phunga no Mageba siyazithoba siyazehlisa, Laba enibabona ngenhla yibo abasunguli balencwadi . Osesandleni sesinxele u Mr S Mkhize kanti osesandleni sokudla u Nkosikazi wakhe u Mrs N Mkhize. Sesifike emaphethelweni ombhalo wethu kanti ukuphetha kwesisodwa isigaba ukuqala kwesinye .

Sifisa ukuniyala Bantwana sinitshele ukuba Ikusasa lenu lisezandleni zenu . Manifisa ukubona ukuba impilo eminyakeni ezayo izonibeka kuphi qalani nibuke izinqumo enizithatha manje nendlela eniziphatha ngayo . Nikhumbule nonke nazalwa nifumbethe kodwa izipho zenu zigcina zishabalala ngenxa yokungayilaleli imithetho yabazali , Impilo iqala ekhaya ngakho ke lalelani imithetho eqhamuka kubantu asebephambili kunani niphinde nifunde ukuhlonipha nibe abantu abanobuntu.

Nisebenze ngokuzikhandla impilo yenu yonke ukufika kulelibanga enikulo , Sekuyisikhathi manje sokuthi nisebenze sengathi usuku lwakusasa alukho ukuze nizovula iminyango ngo Matikuletsheni wenu .

Nikhumbule iminyaka iyasihambela thina bazali kanti sesingadlula emhlabeni nanoma inini , kungakubi mangase sishiye lomhlaba ningazange nasikhombisa impumelelo yenu sisaphila . Sebenzani kanzima ukuze nisenze sishiye lomhlaba ngokukhululeka.

Nikhumbule ukuthi SIYANITHANDA KAKHULU . . .

SINIFISELA OKUHLE KODWA !!!

IMPILO INIPHATHE KAHLE ❤